Extreme Value Modeling of Hydrologic Block Maxima

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Abstract

Accurate estimation of extreme flood magnitudes is central to hydrologic risk assessment, infrastructure design, and long-term water-resources planning. This project investigates stationary extreme-value inference for annual maximum river discharge using likelihood-based methods within the generalized extreme value (GEV) framework. The empirical application focuses on the Potomac River at Little Falls (USGS 01646500), where annual maximum daily discharge is analyzed over a 95-year record of annual block maxima.

The empirical analysis first evaluates the suitability of a stationary modeling framework and then compares the full GEV model with its nested Gumbel submodel. Graphical diagnostics, a Mann--Kendall trend test, and a likelihood-based comparison between stationary and nonstationary GEV specifications provide no evidence against the stationarity assumption. Within this framework, maximum likelihood estimation, quantile--quantile diagnostics, and a likelihood ratio test consistently favor the GEV specification over the constrained Gumbel model. The fitted GEV model yields a positive shape parameter, implying heavier upper-tail behavior and leading to larger return-level estimates, particularly at longer return periods. It also produces substantially wider profile-likelihood confidence intervals, highlighting the practical importance of tail-shape assumptions in flood-frequency analysis.

To assess finite-sample reliability, a Monte Carlo simulation study is conducted under two complementary designs: a parametric GEV benchmark calibrated to the fitted Potomac model and a year-resampling scheme that preserves realistic within-year hydrologic structure. Across record lengths ranging from 30 to 150 years, the parametric GEV experiment evaluates bias and root mean squared error (RMSE) for both parameter estimates and return levels, together with variability summaries for selected parameters and return levels and empirical coverage and interval width for return-level inference. The year-resampling experiment focuses on the behavior of return-level

estimates across resampled datasets, along with empirical coverage and interval width for return levels. The results show that estimation error is driven primarily by sampling variability rather than systematic bias, that return-level uncertainty increases with the return period, and that longer records yield substantial gains in precision.

At the same time, the simulation results indicate that nominal 95% profile-likelihood intervals can undercover in moderate samples, even under correct specification, with somewhat weaker calibration under the year-resampling design. Taken together, the empirical and simulation findings support the use of a stationary block-maxima framework with i.i.d. GEV annual maxima for modeling annual flood maxima at the Potomac River site, while emphasizing that long-horizon return-level inference remains sensitive to tail specification and finite-sample uncertainty.

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Chapter 1

Introduction

Extreme hydrologic events are central to flood-risk assessment, water-resources planning, and the design of infrastructure such as bridges, levees, dams, and urban drainage systems. In these settings, the primary concern is not average river behavior, but the magnitude of extreme and potentially damaging discharge events. Statistical extreme value analysis provides a framework for using limited historical observations to make quantitative statements about such extremes and the uncertainty associated with them.

A standard summary of flood risk is the T-year return level: the discharge level that, under a specified probabilistic model for annual maxima, is exceeded with probability $1/T$ in any given year. Return levels for moderate and long return periods play an important role in hydrologic practice because they translate fitted probability models into quantities that are directly interpretable for design and risk assessment. At the same time, they are inherently difficult to estimate. High return levels lie at or beyond the upper edge of the observed record, so inference must rely not only on the available data but also on assumptions about tail behavior and on the finite-sample stability of the estimation procedure.

This project studies return-level inference for annual maximum river discharge under a stationary block-maxima framework, with particular emphasis on the generalized extreme value (GEV) model and its nested Gumbel special case. The empirical application is based on annual maximum daily discharge for the Potomac River at Little Falls (USGS 01646500), but the broader objective is methodological as well as applied: to clarify how tail-shape specification and finite record length jointly influence extreme quantile estimation in a hydrologic setting.

The central statistical issue is that tail specification matters most where the analysis is asked to extrapolate furthest. The GEV family includes a shape parameter that controls upper-tail behavior, while the Gumbel model imposes the special case of zero shape. When the true tail is heavier than Gumbel, fitting a constrained light-tailed model can materially understate high return levels. Conversely, allowing a freely estimated shape parameter introduces additional sampling variability, especially in moderate samples. For that reason, the choice between a full GEV model and its Gumbel submodel is not merely a technical distinction; it has direct consequences for estimated flood risk and for the width and interpretation of uncertainty intervals.

A second issue concerns record length. Hydrologic monitoring series are finite, and even a relatively long observational history typically contains only a modest number of annual maxima. Classical asymptotic theory helps explain how likelihood-based estimators should behave as sample size increases, but practical inference must still contend with moderate-sample variability. This is especially true for long return periods, where the analysis depends more heavily on the fitted tail model than on direct empirical support from observed extremes. A complete study of return-level inference should therefore address both the empirical fit of the model to the available data and the finite-sample behavior of the inferential tools used to estimate high quantiles.

These considerations motivate the two main components of the project. The first is an empirical analysis of the Potomac River annual-maxima series under a stationary extreme value framework. That analysis addresses whether stationarity is a reasonable working assumption for the observed record, whether the data support a nonzero tail-shape parameter, and how return-level estimates differ between the GEV and Gumbel specifications. The second component is a Monte Carlo simulation study designed to evaluate the finite-sample performance of GEV-based inference under a stationary block-maxima framework across a range of record lengths relevant to hydrologic applications. Together, these two components provide both a site-specific analysis and a broader methodological assessment of how the inference behaves when data are limited.

The project is organized around three guiding questions. First, is a stationary annual-block extreme value model appropriate for the Potomac River annual maxima? Second, within that framework, do the data support the unrestricted GEV model or the nested Gumbel alternative? Third, at realistic record lengths, how reliable are the resulting point estimates and profile-likelihood uncertainty summaries for high return levels? These questions connect directly to the main themes of the project: stationarity, tail-model choice, and finite-sample performance.

The empirical analysis, presented in Chapter 3, examines a series of $n = 95$ annual maximum daily discharge observations. After constructing the annual block-maxima series from the daily record, the chapter documents key data features, including completeness and seasonality, and then assesses whether a stationary modeling framework is reasonable using both nonparametric and likelihood-based evidence. It next fits GEV and Gumbel models under the stationary block-maxima framework by maximum likelihood, compares them through parameter estimates, diagnostics, and a likelihood ratio test, and studies how their return-level estimates and profile-likelihood confidence intervals differ across return periods. The role of that chapter is primarily empirical: it establishes what the Potomac record itself suggests about stationarity, tail behavior, and practical return-level inference.

The simulation study, presented in Chapter 4, complements the observed-data analysis by examining how the same inferential framework behaves across a grid of record lengths. One experiment uses parametric GEV simulation calibrated to the fitted Potomac model, allowing direct measurement of bias, root mean squared error (RMSE), empirical coverage, and interval width relative to known truth. A second experiment resamples complete calendar years from the observed daily discharge record before recomputing annual maxima. This design preserves realistic within-year dependence and seasonality while avoiding the need to specify a fully parametric daily-scale model. The simulation chapter therefore extends the empirical analysis by showing not only what the fitted model estimates for the Potomac record, but also how stable that inferential procedure is likely to be in comparable or shorter records.

The contribution of the project is thus twofold. On the applied side, it provides a coherent likelihood-based analysis of annual flood maxima for a major river gauge and demonstrates the practical effect of tail-shape choice on high return levels. On the methodological side, it gives a structured finite-sample evaluation of the same return-level procedure used in the empirical chapter, including both idealized parametric simulation and a more realistic year-resampling robustness experiment. By combining these components, the project offers a more transparent account of both the strengths and the limitations of stationary block-maxima inference for hydrologic extremes.

The remainder of the project is organized as follows. Chapter 2 develops the theoretical background for the block-maxima framework, the GEV and Gumbel models, return levels, likelihood-based inference, and the Monte Carlo performance measures used later in the simulation study. Chapter 3 presents the stationary extreme value analysis of the Potomac River annual maxima. Chapter 4

reports the simulation study of finite-sample behavior and robustness. Chapter 5 concludes with a synthesis of the main findings, practical implications, and directions for further work.

Chapter 2

Preliminaries

This chapter develops the theoretical and inferential framework used in the stationary analysis of annual maximum discharge in Chapter 3 and in the finite-sample simulation study in Chapter 4. It introduces the annual block-maxima representation, states the working stationarity assumptions, reviews the classical limit theory that motivates the generalized extreme value (GEV) family, and summarizes the likelihood-based tools used later for estimation, model comparison, and uncertainty quantification. It also defines the Monte Carlo performance measures used to evaluate finite-sample behavior in the simulation study. Standard references for these topics include Fisher and Tippett [3], Gnedenko [4], Coles [1], Embrechts, Klüppelberg, and Mikosch [2], Smith [8], van der Vaart [9], Pawitan [7], Mann [6], and Kendall [5].

Throughout this project, inference is conducted under a stationary annual-block framework unless explicitly stated otherwise. Once the daily discharge record is reduced to annual maxima, the resulting sequence is modeled as an approximately independent and identically distributed sample from a common extreme-value distribution. This assumption is adopted as a working modeling framework rather than as a fully testable condition. In particular, the independence assumption is not formally tested, but is considered reasonable at the annual scale because the block-maxima construction reduces short-range dependence present in the daily record.

Chapter 3 focuses on assessing the identical-distribution (stationarity) component of this assumption using graphical diagnostics, a Mann--Kendall trend test, and a likelihood-based comparison between stationary and nonstationary GEV specifications before fitting GEV and Gumbel models under the stationary block-maxima framework, while Chapter 4 examines how the same inferential

procedures behave when only finitely many annual maxima are available. The role of the present chapter is therefore conceptual rather than empirical: it provides the common theoretical language for those two applied chapters.

To keep the chapter focused, source-specific details such as the U.S. Geological Survey data record, the construction of the Potomac annual-maxima sample, and the R implementation used for estimation and simulation are documented in Chapters 3 and 4, where they are used. The present chapter instead develops the statistical framework needed to interpret those analyses.

2.1 Block Maxima, Stationarity, and Trend Assessment

Let $\{Q_t\}_{t\in\mathcal{T}}$ denote a daily discharge time series, measured in cubic feet per second (cfs). Partition the time index into calendar years $Y_1, \ldots, Y_n$, where Y_j contains the daily observations belonging to year j. The annual block maximum for year j is defined by

$$X_j = \max_{t\in Y_j} Q_t, \qquad j = 1, \ldots, n.$$

The sequence $\{X_1, \ldots, X_n\}$ is the annual block-maxima series. This reduction converts a high-frequency hydrologic process into a sample of yearly extremes, which is the natural scale for the return-level questions considered in flood-frequency analysis [1, 2].

In this project, stationarity means that the distribution of annual maxima does not change systematically over time. Operationally, we work under the model

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} G$$

for some fixed distribution function G. The independence assumption is an idealization, but it is more plausible at the annual-maxima level than at the daily scale because the block construction absorbs much of the short-range serial dependence and seasonality present in the original discharge record. The identical-distribution assumption is likewise a working approximation that must be assessed empirically rather than assumed a priori.

For the Potomac application, one component of this assessment is carried out in Chapter 3 using graphical diagnostics and a formal Mann--Kendall trend test. The Mann--Kendall procedure is a rank-based, nonparametric test for monotonic trend that is widely used in hydrology and

environmental time-series analysis because it does not require a specific parametric model for the marginal distribution [6, 5]. Given observations $X_1, \ldots, X_n$, the test statistic is based on the signed pairwise comparisons

$$S = \sum_{1 \le i < j \le n} \operatorname{sgn}(X_j - X_i),$$

where $\operatorname{sgn}(u) = 1$ if $u > 0$, 0 if $u = 0$, and -1 if $u < 0$. Large positive values of S indicate an upward monotonic tendency, whereas large negative values indicate a downward one. Under the null hypothesis of no monotonic trend, a standardized version of S is compared with its null reference distribution to obtain a p-value. The empirical outcome of that test belongs in Chapter 3; here the important point is that stationary extreme-value modeling is justified only after this assumption has been examined at the annual-maxima level.

A complementary assessment of stationarity can be carried out within the GEV model itself by comparing a stationary specification with a nonstationary alternative in which one or more parameters vary over time. In the present project, the model-based comparison used in Chapter 3 contrasts a stationary GEV specification with a nonstationary GEV specification whose location parameter varies linearly with time,

$$\mu_t = \beta_0 + \beta_1 t,$$

while the scale and shape parameters remain constant. Under this formulation, the stationary model corresponds to the null hypothesis $H_0 : \beta_1 = 0$, whereas the alternative $H_1 : \beta_1 \neq 0$ allows a linear temporal trend in location. Because the stationary model is nested within the nonstationary model, the comparison can be carried out using a likelihood ratio test. If ℓ_{stat} and ℓ_{nonstat} denote the maximized log-likelihoods under the stationary and nonstationary models, respectively, then the likelihood ratio statistic is

$$\Lambda_{\text{stat}} = 2\big(\ell_{\text{nonstat}} - \ell_{\text{stat}}\big).$$

Under standard regularity conditions, this statistic is approximately χ^2_1 under H_0, since the comparison introduces one additional parameter. In this way, stationarity can be assessed not only through a distribution-free monotonic-trend test, but also through a model-based comparison directly aligned with the GEV framework used for inference.

The same block-maxima representation underlies both later chapters. In Chapter 3, the observed Potomac annual maxima are modeled directly. In Chapter 4, synthetic annual-maxima

samples are generated either from a fitted GEV distribution under the stationary block-maxima framework or by resampling complete calendar years of daily discharge before recomputing annual maxima. Thus, the block-maxima construction provides the common scale that links the theoretical, empirical, and simulation components of the project.

2.2 Classical Extreme Value Theory

Let $Y_1, Y_2, \ldots$ be independent and identically distributed with common distribution function F, and define the maximum of the first m observations by

$$M_m = \max(Y_1, \ldots, Y_m).$$

Unlike sums, maxima do not generally converge to a nondegenerate limit without normalization. The classical Fisher--Tippett--Gnedenko theorem states that if there exist constants $a_m > 0$ and $b_m \in \mathbb{R}$ such that

$$\frac{M_m - b_m}{a_m} \xrightarrow{d} G$$

as $m \to \infty$, where G is nondegenerate, then G must belong to the GEV family [3, 4, 1]. This theorem provides the fundamental theoretical justification for modeling sufficiently large block maxima by a GEV distribution even when the parent distribution F is unknown.

The theorem is asymptotic rather than exact. It does not assert that annual maxima are exactly GEV-distributed for a finite block size. Rather, it says that the GEV family is the universal nondegenerate limit for normalized maxima whenever such a limit exists. In practice, this gives the GEV model a special status among candidate parametric distributions for block maxima. The empirical question is then whether a GEV model under the stationary block-maxima framework adequately describes the observed annual maxima and, in the present project, whether the additional flexibility of a nonzero tail-shape parameter is needed relative to the nested Gumbel case.

2.3 The Generalized Extreme Value Family

2.3.1 Definition and Tail Interpretation

The generalized extreme value distribution is a three-parameter family with location parameter $\mu \in \mathbb{R}$, scale parameter $\sigma > 0$, and shape parameter $\xi \in \mathbb{R}$. Its cumulative distribution function is

$$G(x; \mu, \sigma, \xi) = \exp\left\{ -\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi} \right\},$$

for values of x satisfying the support condition

$$1 + \xi\left(\frac{x-\mu}{\sigma}\right) > 0.$$

The corresponding density is

$$f(x; \mu, \sigma, \xi) = \frac{1}{\sigma}\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi-1} \exp\left\{ -\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi} \right\}.$$

The shape parameter ξ governs the qualitative behavior of the upper tail [1, 2]:

- if $\xi > 0$, the distribution has a heavy Fréchet-type upper tail;
- if $\xi = 0$, the distribution reduces to the Gumbel case with a lighter, exponential-type tail;
- if $\xi < 0$, the distribution has a finite upper endpoint, corresponding to Weibull-type tail behavior.

This tail interpretation is central to the later chapters. In Chapter 3, the fitted GEV model is compared with the nested Gumbel model to assess whether a nonzero shape parameter is empirically needed for the Potomac annual maxima. In Chapter 4, the same parameter drives the main differences in bias, RMSE, and interval width for longer return periods.

2.3.2 The Gumbel Special Case

The Gumbel model arises as the continuous limit of the GEV family as $\xi \to 0$. Its distribution function is

$$G(x; \mu, \sigma, 0) = \exp\left\{ -\exp\left[-\frac{x-\mu}{\sigma}\right] \right\}.$$

Thus, the Gumbel distribution is not a separate theory but a nested submodel of the GEV family obtained by constraining the shape parameter to zero. Because of this nesting, likelihood-based comparison between the two models is natural and directly interpretable.

The distinction between the unrestricted GEV model and the constrained Gumbel model is substantively important. If ξ is truly nonzero, forcing $\xi = 0$ removes tail flexibility and can induce structural bias in high return levels. If ξ is close to zero, however, the Gumbel model may appear competitive in smaller samples because it avoids the additional variability of estimating the shape parameter. That bias--variance tradeoff is one of the central themes of the simulation study in Chapter 4.

2.4 Return Periods and Return Levels

For a stationary annual-maxima model, the T-year return level z_T is defined as the level exceeded with probability $1/T$ in any given year:

$$\Pr(X > z_T) = \frac{1}{T}.$$

Equivalently,

$$G(z_T; \mu, \sigma, \xi) = 1 - \frac{1}{T}.$$

Solving for z_T yields, for $\xi \neq 0$,

$$z_T = \mu + \frac{\sigma}{\xi}\left[\left\{-\log\left(1 - \frac{1}{T}\right)\right\}^{-\xi} - 1\right],$$

and, for $\xi = 0$,

$$z_T = \mu - \sigma \log\left\{-\log\left(1 - \frac{1}{T}\right)\right\}.$$

These are the return-level formulas used throughout Chapters 3 and 4 [1].

Several qualitative features of z_T are important for interpreting the later results. First, z_T is increasing in T: more extreme return periods correspond to higher quantiles of the annual-maxima distribution. Accordingly, a 100-year return level exceeds a 50-year return level, which in turn exceeds a 10-year return level.

Second, the rate at which z_T increases with T depends strongly on the tail parameter ξ.

Write

$$a_T = -\log\left(1 - \frac{1}{T}\right).$$

For large T, the expansion $a_T \sim 1/T$ implies

$$a_T^{-\xi} \sim T^{\xi}.$$

Hence, for $\xi \neq 0$,

$$z_T \approx \mu + \frac{\sigma}{\xi}(T^{\xi} - 1), \qquad T \to \infty,$$

whereas for the Gumbel case $\xi = 0$,

$$z_T \approx \mu + \sigma \log T, \qquad T \to \infty.$$

Thus, when $\xi > 0$, return levels grow faster than logarithmically and the upper tail becomes increasingly influential as T increases. This is why the GEV and Gumbel return-level curves diverge more clearly at longer return periods and why tail misspecification matters most for long-horizon inference.

A more precise explanation comes from the sensitivity of the return-level function to the underlying GEV parameters. Let

$$a_T = -\log\left(1 - \frac{1}{T}\right), \qquad b_T = a_T^{-\xi}.$$

For $\xi \neq 0$,

$$z_T = \mu + \frac{\sigma}{\xi}(b_T - 1),$$

so that

$$\frac{\partial z_T}{\partial \mu} = 1, \qquad \frac{\partial z_T}{\partial \sigma} = \frac{b_T - 1}{\xi}, \qquad \frac{\partial z_T}{\partial \xi} = \frac{\sigma}{\xi^2}\big[1 - b_T\{1 + \xi \log(a_T)\}\big].$$

In the Gumbel limit $\xi = 0$,

$$z_T = \mu - \sigma \log(a_T), \qquad \frac{\partial z_T}{\partial \mu} = 1, \qquad \frac{\partial z_T}{\partial \sigma} = -\log(a_T).$$

As T increases, a_T decreases toward zero. If $\xi > 0$, then $b_T = a_T^{-\xi}$ grows approximately like T^{ξ}, so

$\partial z_T/\partial\sigma$ is of order T^ξ and $\partial z_T/\partial\xi$ is of order $T^\xi \log T$. Even in the Gumbel case, the derivative with respect to σ grows like $\log T$. These expressions make clear why return-level uncertainty increases with the return period: the mapping from parameter uncertainty to return-level uncertainty is increasingly amplified in the upper tail.

This gradient-based explanation is more informative than a purely verbal appeal to extrapolation. It shows not only that long-period return levels lie farther beyond the observed record, but also why they are statistically more sensitive to uncertainty in σ and especially in ξ. At the same time, the extrapolation interpretation remains substantively important. When the sample contains only n annual maxima, inference for a return period much larger than n relies increasingly on the fitted tail model rather than on direct support from observed extremes. Together, these two perspectives explain why return-level uncertainty increases with T and why tail misspecification is most consequential for large return periods.

2.5 Likelihood-Based Estimation

Let $\theta = (\mu, \sigma, \xi)$ denote the GEV parameter vector and suppose the observed annual maxima are $x_1, \ldots, x_n$. Define

$$y_i(\theta) = 1 + \xi\left(\frac{x_i - \mu}{\sigma}\right), \qquad i = 1, \ldots, n,$$

with the requirement $y_i(\theta) > 0$ for all i. For $\xi \neq 0$, the GEV log-likelihood is

$$\ell(\theta) = -n\log\sigma - \left(1 + \frac{1}{\xi}\right)\sum_{i=1}^{n}\log y_i(\theta) - \sum_{i=1}^{n} y_i(\theta)^{-1/\xi}.$$

The maximum likelihood estimator is

$$\hat{\theta} = \arg\max_{\theta} \ell(\theta).$$

Maximum likelihood is the estimation principle used for both the GEV and Gumbel models in the observed-data analysis and throughout the simulation study. Its appeal lies in invariance and large-sample efficiency, although GEV likelihood theory requires care because the support depends on the parameters [8, 1].

Under standard regularity conditions and away from problematic boundary cases, maximum

likelihood estimators are asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\big(0, I_E(\theta_0)^{-1}\big),$$

where $I_E(\theta_0)$ is the expected Fisher information matrix at the true parameter value θ_0 [8, 9]. This asymptotic approximation is not treated as a substitute for finite-sample assessment in this project; rather, it provides the theoretical benchmark against which the simulation results are interpreted.

2.6 Likelihood Ratio Testing and Model Comparison

Because the Gumbel model is nested within the GEV family through the constraint $\xi = 0$, the two models can be compared using a likelihood ratio test. Let ℓ_{GEV} and ℓ_{Gumbel} denote the maximized log-likelihood values under the unrestricted and constrained models, respectively. The likelihood ratio statistic is

$$\Lambda = 2\big(\ell_{\text{GEV}} - \ell_{\text{Gumbel}}\big).$$

Under the null hypothesis $H_0 : \xi = 0$, and under standard regularity conditions, Λ is approximately distributed as χ^2_1 [7, 9].

This model comparison has two later uses. In Chapter 3, it provides a formal test of whether allowing a nonzero shape parameter materially improves fit for the Potomac annual maxima. In Chapter 4, the same nested relationship clarifies the practical consequences of tail misspecification by comparing a correctly specified GEV fit with a constrained Gumbel fit under GEV truth.

2.7 Profile Likelihood and Likelihood-Based Uncertainty

For a scalar parameter or function of interest $\psi = \psi(\theta)$, the profile log-likelihood is

$$\ell_p(\psi) = \max_{\theta:\, \psi(\theta)=\psi} \ell(\theta).$$

A nominal $100(1-\alpha)\%$ profile-likelihood confidence interval is then obtained by inverting the likelihood ratio criterion

$$2\{\ell(\hat{\theta}) - \ell_p(\psi)\} \leq \chi^2_{1,1-\alpha}.$$

Compared with Wald intervals based solely on asymptotic normality, profile-likelihood intervals often behave better in moderate samples and for nonlinear functions of the parameters because they follow the shape of the likelihood surface directly [7, 1].

Profile likelihood is used later for two distinct inferential purposes. In Chapter 3, it is used to quantify uncertainty in the shape parameter and in return levels. In Chapter 4, the same interval construction is applied repeatedly across Monte Carlo replications so that empirical coverage and mean interval width can be evaluated for the very procedure used in the observed-data analysis. That connection is important: the simulation study is not assessing an abstract notion of interval uncertainty, but the specific likelihood-based procedure adopted for the Potomac application.

2.8 Delta Method for Return-Level Variance

Return levels are smooth functions of the GEV parameter vector, so their sampling variability can also be studied through the multivariate delta method. Let

$$z_T = g_T(\theta).$$

Then, under the same asymptotic normality used for the MLE,

$$\sqrt{n}\big(\hat{z}_T - z_T(\theta_0)\big) \xrightarrow{d} N\big(0, \nabla g_T(\theta_0)^\top I_E(\theta_0)^{-1} \nabla g_T(\theta_0)\big)\,,$$

so that

$$\mathrm{Var}(\hat{z}_T) \approx \frac{1}{n}\,\nabla g_T(\theta_0)^\top I_E(\theta_0)^{-1} \nabla g_T(\theta_0).$$

This formula gives the baseline variance decay $\mathrm{Var}(\hat{z}_T) = O(n^{-1})$ for fixed T, and hence standard errors of order $O(n^{-1/2})$ [9]. This asymptotic contraction is the theoretical counterpart of the decreasing RMSE and the reduction in profile-likelihood interval width as record length increases in the simulation study.

The most important qualitative implication, however, is not just the role of n but also the role of T. Because the gradient of g_T grows with the return period, the transformation from parameter uncertainty to return-level uncertainty becomes progressively more severe as one moves farther into the upper tail. This provides a theoretical explanation for the pattern observed in the

simulation study, where both RMSE and profile-likelihood interval width are larger for the 50-year and 100-year return levels than for the 10-year return level, reflecting the increasing sensitivity of z_T to the model parameters as T increases.

2.9 Monte Carlo Performance Measures

Chapter 4 evaluates finite-sample performance through Monte Carlo replication. For each record length n and each data-generating mechanism, let $\hat{\psi}^{(b)}$ denote the estimate of a target quantity ψ_0 from replication b, where $b = 1, \ldots, B$.

2.9.1 Bias and Root Mean Squared Error

Monte Carlo bias is defined by

$$\mathrm{Bias}(\hat{\psi}) = \frac{1}{B}\sum_{b=1}^{B}\big(\hat{\psi}^{(b)} - \psi_0\big),$$

and Monte Carlo root mean squared error (RMSE) is

$$\mathrm{RMSE}(\hat{\psi}) = \left[\frac{1}{B}\sum_{b=1}^{B}\big(\hat{\psi}^{(b)} - \psi_0\big)^2\right]^{1/2}.$$

Bias measures systematic deviation from the target, whereas RMSE combines bias and variability into a single measure of overall estimation accuracy. These summaries are used in Chapter 4 for both parameter estimators and return-level estimators under the parametric GEV data-generating mechanism.

2.9.2 Empirical Coverage and Interval Width

If $[L_T^{(b)}, U_T^{(b)}]$ denotes a nominal 95% confidence interval for the T-year return level in replication b, then empirical coverage is estimated by

$$\widehat{\mathrm{Cov}}_n(T) = \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\big\{L_T^{(b)} \le z_T(\theta_0) \le U_T^{(b)}\big\},$$

and mean interval width is estimated by

$$\widehat{\mathrm{Width}}_n(T) = \frac{1}{B}\sum_{b=1}^{B}\big(U_T^{(b)} - L_T^{(b)}\big).$$

Under the parametric GEV experiment of Chapter 4, the target $z_T(\theta_0)$ is known exactly, so empirical coverage has its usual literal interpretation. Under the year-resampling experiment, the same target is used as a reference quantity to maintain comparability across designs; in that setting, the resulting coverage summaries are best interpreted as reference-based robustness diagnostics rather than exact coverage relative to a known annual-maxima truth.

These performance measures connect the theory to the practical goals of the project. Bias and RMSE address whether GEV fitting under the stationary block-maxima framework yields stable point estimates at record lengths comparable to the Potomac series. Empirical coverage and interval width address whether the uncertainty statements attached to those estimates are well calibrated and how much they improve as additional years of data become available.

2.10 Summary

The preliminaries developed here provide the conceptual bridge between the observed-data analysis and the simulation study. The block-maxima construction defines the data scale of interest, the Mann--Kendall test and likelihood-based stationarity comparison provide formal frameworks for assessing the stationary assumption before stationary modeling is adopted, the Fisher--Tippett--Gnedenko theorem motivates the GEV family, and the nested Gumbel case provides a natural benchmark for model comparison. Return levels translate fitted extreme-value models into interpretable flood-frequency quantities, while likelihood-based methods provide the tools for estimation, model comparison, and uncertainty quantification. Finally, the delta-method representation of return-level variance and the Monte Carlo performance measures establish the theoretical basis for the finite-sample results reported in Chapter 4.

Chapters 3 and 4 can therefore be read as two complementary applications of the same framework: the first to the observed Potomac annual maxima, and the second to repeated synthetic datasets used to evaluate the finite-sample behavior of that inferential procedure.

Chapter 3

Results of Stationary Extreme Value Analysis

This chapter presents the stationary extreme value analysis of annual maximum mean daily discharge for the Potomac River at the Little Falls gauge (USGS 01646500). The objective is to determine whether a stationary block-maxima model provides an adequate description of the observed yearly extremes and, within that stationary framework, whether the additional flexibility of the generalized extreme value (GEV) model is warranted relative to the nested Gumbel specification. The analysis therefore proceeds from data construction and preliminary data checks to stationarity assessment, model fitting, diagnostic comparison, formal model comparison, and return-level estimation.

To avoid repeating material already developed in Chapter 2, this chapter emphasizes empirical findings rather than general theory. Definitions of the block-maxima framework, the GEV and Gumbel families, return periods, return levels, and likelihood-based interval construction were given previously. Here, the focus is on what those methods reveal for the Potomac River record.

3.1 Annual Block Maxima Series

Daily mean discharge observations were aggregated to calendar-year maxima, yielding the annual block-maxima series used throughout the chapter. Under this construction, the analysis

dataset contains $n = 95$ annual maxima spanning 1930--2024. The resulting sample provides the empirical basis for all model fitting, diagnostics, and return-level calculations reported below.

Because the statistical analysis is built on annual maxima, it is useful to see how these block maxima are obtained from the daily record. Figure 3.1 shows the full daily mean discharge series together with the annual maxima highlighted. The daily record exhibits large short-term variation, but the block-maxima step reduces each calendar year to its single most extreme daily observation. This is precisely the data reduction required by the classical block-maxima extreme value framework introduced in Chapter 2. The annual-maxima series is therefore not an auxiliary summary, but the fundamental dataset on which the GEV and Gumbel analyses under the stationary block-maxima framework are based.

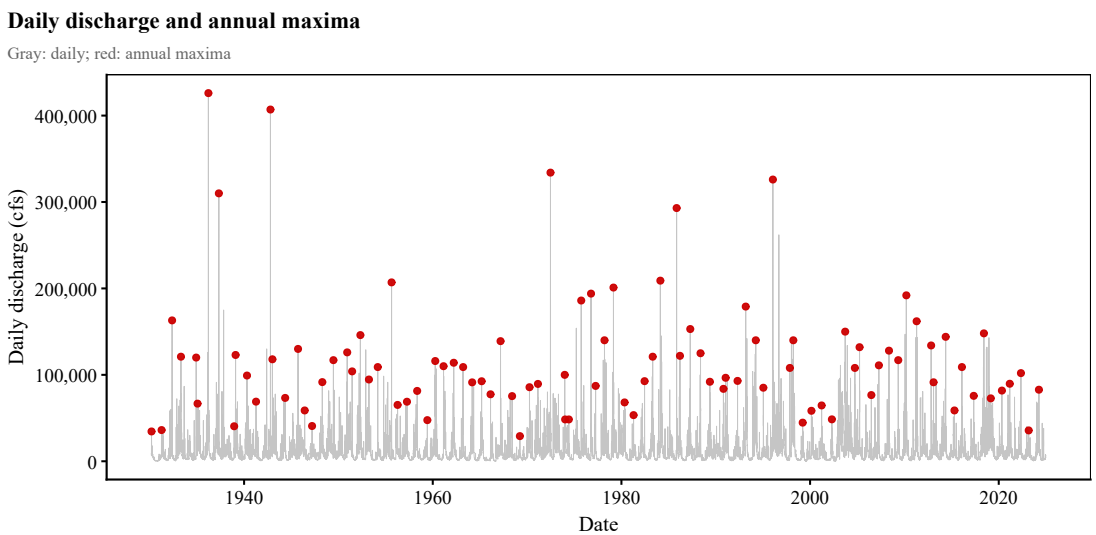

Figure 3.1: Daily mean discharge for the Potomac River at Little Falls, with the annual maximum in each year highlighted. The stationary extreme value analysis is based on these annual block maxima rather than on the full daily record.

3.2 Data Checks: Completeness and Seasonality

Before fitting stationary extreme value models, it is important to verify that the annual-maxima series is derived from a sufficiently complete daily record and to understand the seasonal context in which the yearly extremes occur. In the available data, there are no missing daily discharge values within the observed record used for analysis. The year 1930 is partial because the daily series begins

in March, but this reflects the beginning of measurement rather than missing observations after the start of the record. Thus, the annual maxima analyzed here are derived from complete daily information within the available measurement period.

Seasonality is also clearly present in the daily discharge process. The left panel of Figure 3.2 shows the month in which each annual maximum occurs, while the right panel summarizes the distribution of monthly maxima across the calendar year. Annual maxima cluster primarily in late winter and spring, and the largest monthly maxima are concentrated in these same months. The monthly-maxima distributions are also more variable during that part of the year, indicating that the timing of extreme discharge is not uniform across seasons.

These patterns matter for interpretation, but they do not invalidate the block-maxima approach. Rather, they clarify what the annual-maxima analysis is and is not doing. The block-maxima framework does not model seasonal variation explicitly; instead, it summarizes each year by its most extreme observed daily discharge, regardless of when within the year that event occurs. Consequently, the inference in this chapter is conducted at the yearly scale. Seasonal structure is present in the underlying daily process, but the statistical model for annual maxima treats each year as contributing one extreme observation.

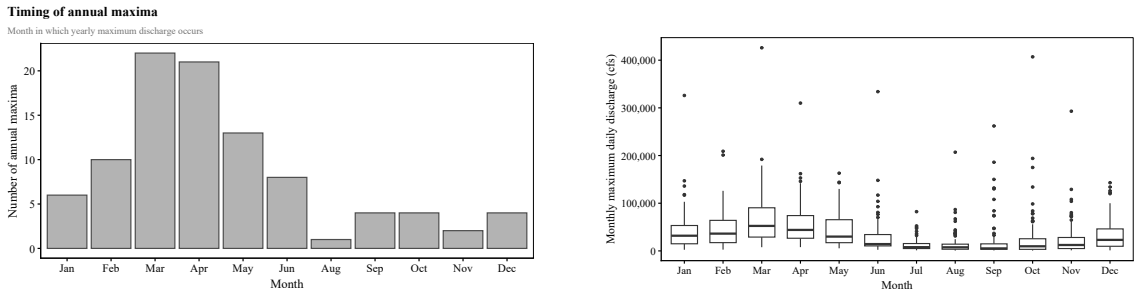

Figure 3.2: Seasonal structure in the Potomac daily discharge record. Left: months in which annual maxima occur. Right: boxplots of monthly maximum daily discharge. Annual maxima cluster primarily in late winter and spring, and extremes are larger and more variable during these months.

3.3 Assessment of Stationarity

The first modeling question is whether the annual maxima exhibit evidence of systematic temporal change that would argue against a stationary model. Figure 3.3 shows the annual maxima plotted against year together with a least-squares reference line. The series displays substantial

interannual variability, including several unusually large flood years, but no persistent upward or downward pattern is visually apparent. The fitted line is nearly flat, which is consistent with the absence of an obvious long-run trend in the magnitude of annual extremes.

This graphical impression is reinforced by the Mann--Kendall test applied to the annual-maxima sequence. The test produced a p-value of $p = 0.918$, so there is no statistically significant evidence of a monotonic trend at the 5% level. On its own, this result supports the use of a stationary annual-block framework as a reasonable working assumption for the Potomac annual maxima.

Because stationarity is central to the subsequent modeling, the analysis also included a model-based comparison between a stationary GEV specification and a nonstationary GEV specification with linear trend in the location parameter. That likelihood ratio test produced a p-value of $p = 0.263$, providing no evidence that a linear time trend in location improves the fit. Thus, both the nonparametric trend assessment and the likelihood-based model comparison point to the same conclusion: there is no evidence against stationarity at the annual scale for this record.

Accordingly, the subsequent analyses proceed under GEV and Gumbel models within the stationary block-maxima framework. This does not imply that the hydrologic process is fully invariant over time, only that the available annual-maxima record does not show evidence strong enough to reject a stationary annual-block framework for the purposes of the present analysis.

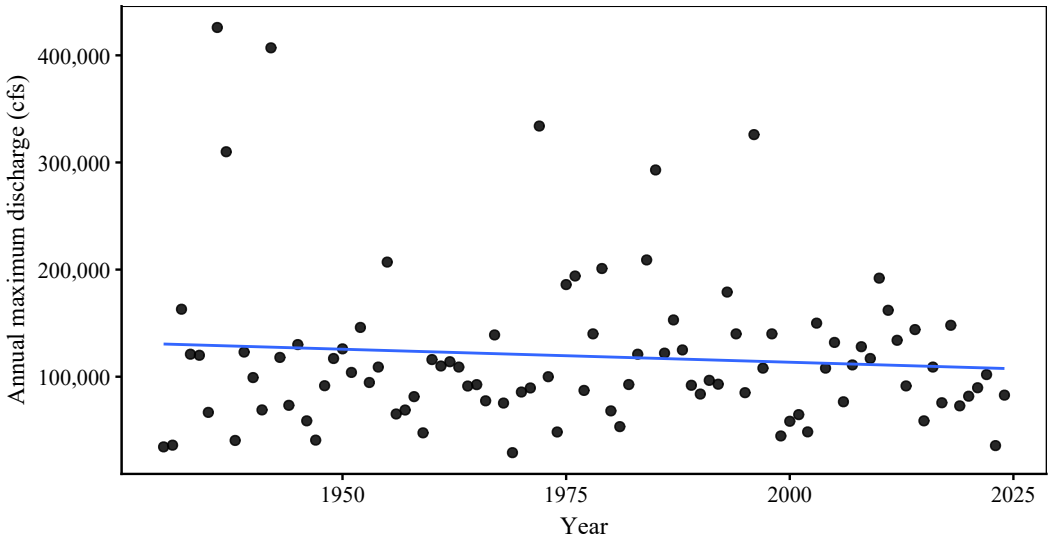

Figure 3.3: Annual maximum discharge for the Potomac River at the Little Falls gauge (USGS 01646500) with a fitted linear trend line for visual reference.

3.4 Stationary Model Fits

Under the stationary framework supported by the preceding analysis, two nested extreme value models were fitted by maximum likelihood: an unrestricted GEV model and the Gumbel submodel obtained by fixing the shape parameter at $\xi = 0$. In the computational workflow, point estimates and standard errors for these model parameters were obtained from the `ismev` package, while profile-likelihood interval calculations were obtained separately from `extRemes` for quantities where likelihood-based uncertainty summaries were needed.

Table 3.1 reports the maximum likelihood estimates and associated standard errors. The estimated location and scale parameters are broadly similar under the two specifications, so the principal difference between the models lies in the fitted tail behavior. Under the GEV model, the estimated shape parameter is positive, $\hat{\xi} = 0.1845$ with standard error 0.0877, indicating heavier upper-tail behavior than is allowed under the Gumbel model. This is the key substantive feature of the fitted GEV model, because a positive shape parameter implies more mass in the far upper tail and therefore larger extrapolated flood quantiles at long return periods.

A simple Wald calculation based on Table 3.1 gives an approximate 95% confidence interval of (0.013, 0.356) for ξ, which excludes zero. Because the shape parameter is central to the model comparison, it is more informative to look directly at the likelihood-based interval computed from the `extRemes` fit. The corresponding 95% profile-likelihood interval is (0.0625, 0.3260), which also excludes zero. Thus, both the normal approximation and the likelihood-based interval point in the same direction: the observed annual maxima support a positive tail-shape parameter rather than the lighter-tailed Gumbel restriction.

Table 3.1: Maximum likelihood estimates (standard errors) for GEV and Gumbel models fit to annual maximum discharge under the stationary block-maxima framework.

Model	Parameter	Estimate (SE)
GEV	μ	85,818 (9,814)
GEV	σ	41,484 (5,337)
GEV	ξ	0.1845 (0.0877)
Gumbel	μ	90,274 (3,013)
Gumbel	σ	45,328 (2,131)

3.5 Diagnostic Comparison of GEV and Gumbel Fits

Model adequacy was examined using quantile--quantile (Q--Q) plots comparing the empirical quantiles of the annual maxima with the corresponding theoretical quantiles from each fitted model. In the analysis code, these theoretical quantiles were computed directly from the `ismev` maximum likelihood fits, so the diagnostic plots are fully aligned with the parameter estimates in Table 3.1.

Figure 3.4 shows that the fitted GEV model tracks the empirical quantiles closely across most of the observed range, including the upper tail. The Gumbel fit is adequate through the central portion of the distribution but departs more noticeably at larger quantiles, where the most extreme observations fall below the pattern implied by the fitted Gumbel model. This indicates that the Gumbel model underestimates upper-tail quantiles relative to the data, which is consistent with the need for a heavier upper tail than the Gumbel distribution can provide.

Color is used to encode time, with cooler colors representing earlier years and warmer colors representing more recent years. The absence of systematic color clustering along or away from the reference line provides additional visual support for the assumption that the marginal distribution of annual maxima is approximately time-invariant over the observation period.

These diagnostics are especially important because the scientific interest lies not merely in fitting the bulk of the annual-maxima distribution, but in representing the upper tail accurately enough for return-level extrapolation. In that respect, the Q--Q plots favor the GEV model under the stationary block-maxima framework. The main gain from allowing ξ to vary is not a substantial change in the center of the distribution, but a materially improved fit to the largest observed discharges.

3.6 Likelihood Ratio Comparison

The graphical comparison is supported by a formal likelihood ratio test of the Gumbel restriction $H_0 : \xi = 0$ against the unrestricted GEV alternative. Using the maximized log-likelihood values from the `ismev` fits, the test statistic is

$$\Lambda = 2\left(\ell_{\mathrm{GEV}} - \ell_{\mathrm{Gumbel}}\right) = 7.932. \tag{3.1}$$

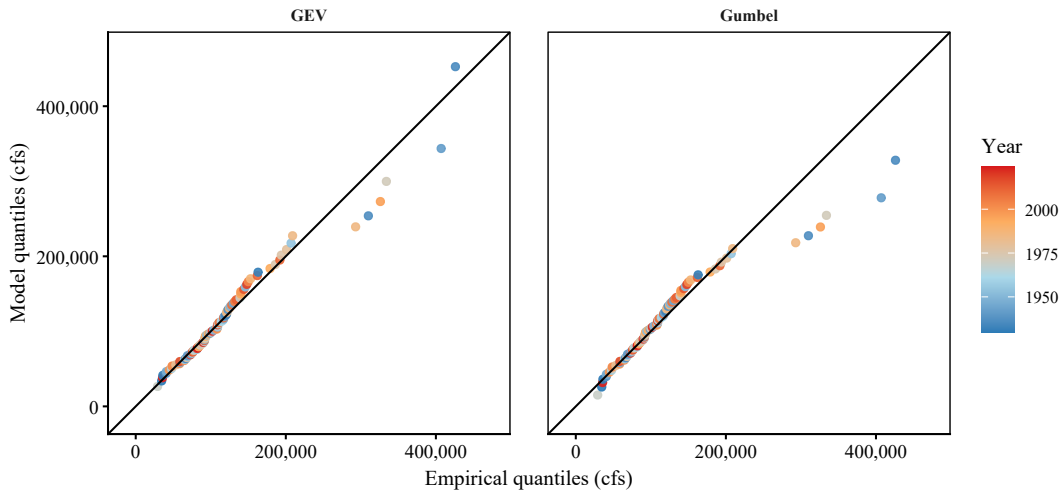

Figure 3.4: Q--Q plots comparing empirical quantiles of annual maximum discharge to theoretical quantiles from fitted GEV and Gumbel models. Color encodes year, with cooler colors indicating earlier observations and warmer colors indicating more recent observations.

Under the standard regularity conditions for this nested comparison, the reference distribution is approximated by χ^2_1, giving a p-value of $p = 0.00486$.

This result provides statistically significant evidence that allowing a nonzero shape parameter improves model fit. The likelihood ratio test therefore reaches the same conclusion as the fitted value of $\hat{\xi}$, the profile-likelihood interval for ξ, and the Q--Q diagnostics: the GEV model under the stationary block-maxima framework gives a better description of the Potomac annual maxima than the nested Gumbel model. For the remainder of the chapter, the Gumbel fit is still useful as a benchmark, but the unrestricted GEV specification is the preferred model for inference on extreme discharge levels.

3.7 Return-Level Results

Return levels were estimated for return periods $T = 5, 10, \ldots, 200$ years under both fitted stationary models. In the computational workflow, return-level point estimates and 95% profile-likelihood confidence intervals were obtained from `extRemes::fevd` and `extRemes::ci.fevd` using `method = "proflik"`. For the real-data analysis, the profile likelihood for each return period was evaluated numerically on a grid with $n_{\text{int}} = 400$ points over a data-adaptive candidate range.

This detail is worth recording here because it is part of the actual analysis pipeline used to produce the reported intervals.

Figure 3.5 displays the fitted return-level curves and their 95% profile-likelihood confidence intervals. At shorter return periods, the GEV and Gumbel estimates are fairly close, which is consistent with the fact that both models fit the central portion of the annual-maxima distribution reasonably well. As the return period increases, however, the two fitted curves diverge. The GEV model produces increasingly larger return-level estimates than the Gumbel model, and that separation becomes substantial in the long-return range. This pattern is exactly what one would expect from the positive fitted shape parameter in the GEV model: once extrapolation moves farther into the upper tail, tail-shape assumptions become increasingly influential.

The uncertainty summaries in Figure 3.6 tell the same story from a second perspective. For both models, confidence intervals widen as the return period grows, reflecting the increasing difficulty of estimating quantiles that lie well beyond the observed 95-year record. At relatively short return periods, the GEV intervals are comparable to, and in the smallest cases even slightly narrower than, the corresponding Gumbel intervals. But as T increases, the GEV intervals widen much more rapidly and eventually become substantially larger than the Gumbel intervals. This widening reflects the fact that uncertainty about the tail-shape parameter has little effect on modest extrapolations but a large effect on long-horizon return levels.

To understand this interval behavior more directly, it is useful to examine the profile likelihood as a function of the return level itself. Figure 3.7 shows the profile-likelihood curves for the 50-year and 100-year return levels under both fitted models. The confidence intervals are obtained by inverting the likelihood ratio criterion, so their width is governed by the curvature of the profile likelihood near its maximum. In these plots, the GEV profile is flatter than the Gumbel profile near the maximizer, especially at the 100-year return level, so the likelihood-ratio cutoff is reached farther away from the estimated return level. This produces wider profile-likelihood intervals under the GEV model. The same plots also show that the Gumbel profile is more asymmetric, which helps explain why the Gumbel intervals can appear more skewed even when they are narrower overall. Thus, the profile-likelihood figures provide a direct geometric explanation of the interval patterns seen in Figures 3.5 and 3.6.

Table 3.2 provides numerical summaries for representative return periods. The table confirms the graphical comparison. For example, the estimated 100-year return level is markedly higher under

the GEV model than under the Gumbel model, and its 95% profile-likelihood interval is also much wider. This is an important substantive finding rather than a technical detail: for this dataset, the simpler Gumbel model would understate both the magnitude of extreme flood discharges and the uncertainty attached to those extrapolated levels.

Taken together, the evidence from parameter estimation, likelihood-based model comparison, Q--Q diagnostics, and return-level extrapolation is internally consistent. The annual maxima do not show evidence of a temporal trend strong enough to reject a stationary framework, but within that stationary framework the data favor a positively shaped GEV tail over the Gumbel restriction. The practical consequence is that high-end flood quantiles for the Potomac River at Little Falls are estimated to be both larger and more uncertain than they would appear under a lighter-tailed Gumbel model. For this reason, the GEV model under the stationary block-maxima framework is adopted as the primary model for subsequent interpretation and for the simulation study in Chapter 4.

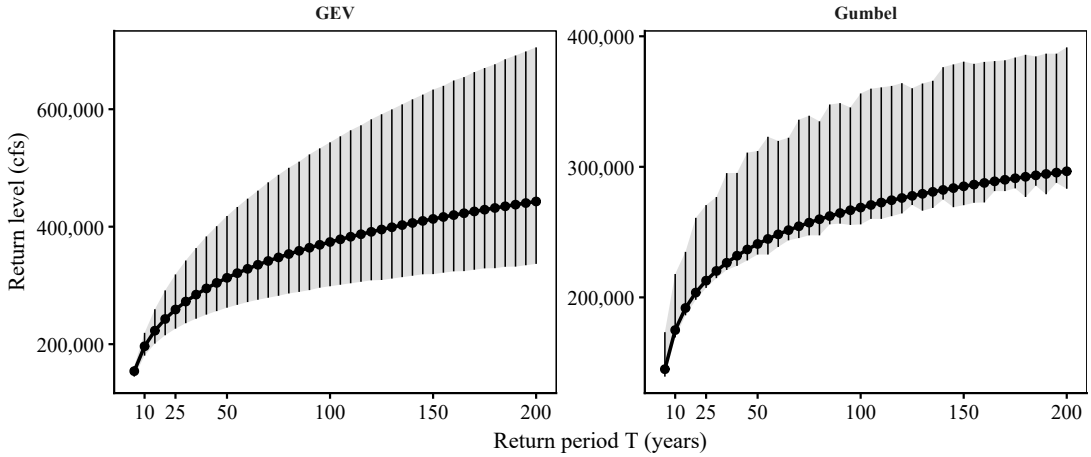

Figure 3.5: Estimated return levels with 95% profile-likelihood confidence intervals for annual maximum discharge under fitted GEV and Gumbel models within the stationary block-maxima framework.

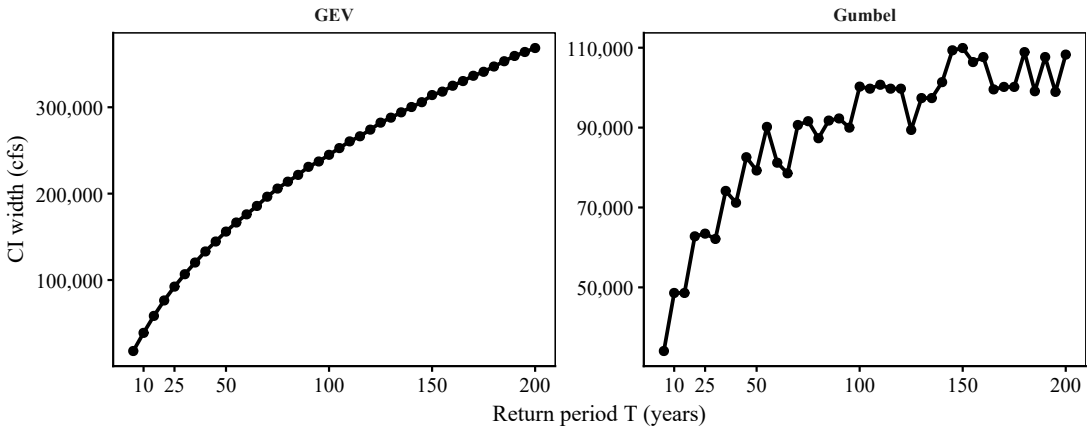

Figure 3.6: Width of 95% profile-likelihood confidence intervals for return levels under fitted GEV and Gumbel models within the stationary block-maxima framework.

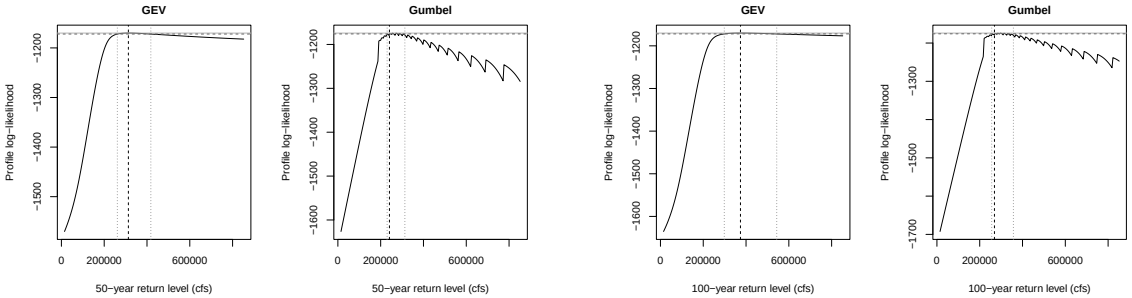

Figure 3.7: Profile-likelihood curves for the 50-year and 100-year return levels under the fitted GEV and Gumbel models within the stationary block-maxima framework. Flatter curvature near the maximum corresponds to wider profile-likelihood confidence intervals, and the difference between the two models becomes more pronounced at longer return periods.

Table 3.2: Selected return-level estimates and 95% profile-likelihood confidence intervals.

Model	Return period	Return level	95% CI	CI width
GEV	5	154,063	[145,007, 161,840]	16,833
Gumbel	5	144,992	[140,799, 173,788]	32,990
GEV	10	196,399	[180,534, 219,382]	38,847
Gumbel	10	174,962	[170,257, 219,234]	48,977
GEV	25	258,958	[226,677, 318,456]	91,779
Gumbel	25	212,830	[208,131, 270,621]	62,490
GEV	50	312,942	[261,676, 418,144]	156,468
Gumbel	50	240,922	[234,523, 309,913]	75,390
GEV	100	373,894	[297,986, 543,759]	245,773
Gumbel	100	268,807	[258,631, 351,002]	92,371

Chapter 4

Simulation Study for GEV Return-Level Inference

This chapter investigates the finite-sample behavior of the stationary generalized extreme value (GEV) procedures used in Chapter 3. Whereas the previous chapter analyzed the observed Potomac River record and identified the GEV model under the stationary block-maxima framework as the preferred description of the annual maxima, the present chapter addresses a complementary methodological question: how reliable are GEV-based point estimates and profile-likelihood uncertainty statements when inference is based on only a moderate number of annual maxima? This question is central in hydrologic applications, where even relatively long monitoring programs often yield only a few decades of annual extremes and where the quantities of greatest interest, such as 50- or 100-year return levels, require extrapolation into the upper tail.

The simulation study is designed to evaluate this question over a range of record lengths that are short, moderate, and long relative to the Potomac series. In the real-data analysis, inference was based on $n = 95$ annual maxima. Here, record length varies over

$$n \in \{30, 50, 75, 95, 150\},$$

so that the observed Potomac record can be interpreted within a broader finite-sample spectrum. For each record length, $B = 500$ Monte Carlo datasets are generated and analyzed.

Two complementary simulation designs are considered. The primary experiment is a parametric benchmark in which annual maxima are generated directly from a GEV distribution under a stationary block-maxima framework calibrated to the fitted Potomac model. This design provides a correct-specification setting in which bias, root mean squared error (RMSE), and interval behavior can be assessed relative to known truth. The second experiment resamples complete observed calendar years of daily discharge with replacement and then recomputes annual maxima. That design preserves within-year seasonality and short-range hydrologic dependence while removing cross-year dependence through independent year-level resampling. Taken together, the two experiments separate intrinsic finite-sample behavior under idealized GEV truth from robustness under a more realistic daily-scale data-generating mechanism.

The chapter has four empirical aims. First, under parametric GEV truth, it quantifies bias and RMSE for the parameter estimators $(\hat{\mu}, \hat{\sigma}, \hat{\xi})$. Second, it studies point-estimation accuracy for return levels at $T \in \{10, 50, 100\}$ years and examines the sampling distribution of the estimated 100-year return level. Third, it evaluates the empirical coverage and mean width of nominal 95% profile-likelihood confidence intervals. Fourth, it compares correctly specified GEV fits with constrained Gumbel fits in order to assess the practical consequences of tail misspecification.

Because the same inferential pipeline is applied throughout, differences across record lengths or across data-generating mechanisms can be attributed to finite-sample variation and model structure rather than to changes in the estimation procedure. In particular, the simulation code fixes the truth at the GEV maximum likelihood estimates from the Potomac analysis under the stationary block-maxima framework, draws parametric samples with `evd::rgev`, fits simulated annual-maxima samples with `extRemes::fevd`, and computes return levels and profile-likelihood intervals with the same `extRemes` workflow used for the real-data return-level analysis. For the simulation runs, the profile likelihood is evaluated on a grid of `nint = 200` points, with $B = 500$ replications in both the parametric and year-resampling experiments.

4.1 Simulation Design

The primary experiment uses a GEV data-generating mechanism under a stationary block-maxima framework. Let

$$\theta_0 = (\mu_0, \sigma_0, \xi_0)$$

denote the GEV parameter vector obtained by fitting the stationary Potomac annual-maxima series in Chapter 3. For each record length n and each replication, independent annual maxima are generated from

$$X_1, \ldots, X_n \sim \mathrm{GEV}(\mu_0, \sigma_0, \xi_0).$$

This design provides a correct-specification benchmark: any observed bias in parameter estimates or return levels, and any departure of empirical coverage from the nominal 95% level, is attributable to finite-sample effects rather than to misspecification of the annual-maxima model. Because the fitted Potomac shape parameter is positive, this benchmark is also calibrated to the heavier upper-tail behavior identified in the empirical analysis.

Each simulated annual-maxima sample is then analyzed with the same stationary GEV return-level workflow used in the empirical study. In the simulation code, GEV fits are obtained with `extRemes::fevd` by maximum likelihood, and return levels together with profile-likelihood confidence intervals are computed for $T \in \{10, 50, 100\}$ using the `proflik` option. The simulated samples are also fitted with a constrained stationary Gumbel model. That comparison isolates the effect of imposing $\xi = 0$ when the true tail shape is positive.

The second experiment replaces the parametric annual-maxima model with a resampling design built from the observed daily discharge record. Let $\mathcal{Y}_1, \ldots, \mathcal{Y}_m$ denote the complete calendar years in the daily series, where $\mathcal{Y}_j$ contains all daily observations from year j. For each replication and record length n, years are sampled independently with replacement from $\{1, \ldots, m\}$, and an annual maximum is computed from each selected year. Thus, each synthetic dataset again consists of n annual maxima, but those maxima are induced by the empirical daily process rather than by direct GEV simulation.

This year-resampling design preserves several realistic features of the daily record within each sampled year, including the seasonal cycle, within-year dependence, and the marginal distribution of daily discharge. At the same time, independent resampling across years breaks any cross-year dependence. The goal is not to define a new exact annual-maxima truth, but to assess whether the qualitative patterns in variability, empirical coverage, and interval width observed in the parametric experiment remain evident when the block maxima arise from a more realistic hydrologic structure.

4.2 Performance Summaries

The main Monte Carlo summaries are bias, RMSE, empirical coverage, and mean interval width. Under parametric GEV truth, these summaries are evaluated against known targets determined by θ_0. For return period T, the true return level is denoted by $z_T(\theta_0)$.

For parameter estimation, bias and RMSE are reported for $\hat{\mu}$, $\hat{\sigma}$, and $\hat{\xi}$ across the record-length grid. For return-level estimation, the study reports the Monte Carlo mean, bias, and RMSE of $\hat{z}_T$ for $T \in \{10, 50, 100\}$. The distribution of the estimated 100-year return level is also examined graphically, because scalar summaries alone do not fully reveal skewness or the occurrence of large upper realizations.

For interval estimation, the focus is on nominal 95% profile-likelihood confidence intervals. Under parametric GEV truth, empirical coverage is the proportion of intervals that contain $z_T(\theta_0)$, and mean width measures average interval precision. Under year-resampling, the same reference return level $z_T(\theta_0)$ is retained so that results remain comparable with the parametric experiment. In that setting, the reported coverages should be interpreted as reference-based robustness diagnostics rather than exact coverage probabilities under a known annual-maxima law.

Theoretical definitions of return levels, likelihood-based inference, and the Monte Carlo criteria themselves were developed in Chapter 2. Accordingly, the emphasis here is on what the simulation shows rather than on re-deriving those quantities.

4.3 Parameter Estimation Under Parametric GEV Truth

We begin with the finite-sample behavior of GEV maximum likelihood estimation under correct specification within the stationary block-maxima framework. Table 4.1 reports Monte Carlo bias and RMSE for $(\hat{\mu}, \hat{\sigma}, \hat{\xi})$ across the record-length grid.

Table 4.1: Monte Carlo bias and RMSE of GEV parameter estimators under parametric GEV truth.

n	Bias($\hat{\mu}$)	RMSE($\hat{\mu}$)	Bias($\hat{\sigma}$)	RMSE($\hat{\sigma}$)	Bias($\hat{\xi}$)	RMSE($\hat{\xi}$)
30	-58.82	7965.57	-49.12	7323.30	-0.0040	0.1842
50	513.29	6871.76	-112.93	5630.52	-0.0057	0.1419
75	74.77	5483.42	66.07	4825.26	0.0016	0.1086
95	251.45	4761.75	124.83	4377.23	0.0016	0.0886
150	104.20	3859.51	-158.28	3578.55	-0.0030	0.0656

The first conclusion is that bias is small relative to RMSE for all three parameters. For the

shape parameter, absolute bias never exceeds 0.0057, whereas RMSE declines from 0.1842 at $n = 30$ to 0.0656 at $n = 150$. The same pattern holds for the location and scale parameters. At the Potomac record length $n = 95$, for example, the biases of $\hat{\mu}$ and $\hat{\sigma}$ are 251.45 and 124.83, respectively, while the corresponding RMSE values are 4,761.75 and 4,377.23. In practical terms, the dominant source of estimation error is sampling variability rather than systematic bias. At $n \approx 95$, bias is negligible relative to overall estimation error, but variability remains substantial.

A second conclusion is that precision improves steadily with record length. RMSE decreases monotonically for all three parameters, and the proportional reduction is especially marked for the shape parameter. From $n = 30$ to $n = 150$, RMSE falls by about 52% for both μ and σ, and by about 64% for ξ. Figure 4.1 makes this especially clear for the shape parameter: the sampling distribution of $\hat{\xi}$ contracts visibly as the number of annual maxima increases. Figure 4.2 complements that picture by showing that bias fluctuates around zero without any persistent directional trend.

These results place the empirical Potomac analysis in a useful context. The record length $n = 95$ is plainly more stable than the short-record settings $n = 30$ and $n = 50$, yet it does not behave like an effectively infinite-sample regime. In particular, estimation of the tail-shape parameter remains appreciably variable, even though the simulation indicates that the observed Potomac record is well beyond the most unstable part of the small-sample range. This matters because, as shown in Chapter 3, the fitted shape parameter is precisely the feature that separates the preferred GEV analysis from the Gumbel alternative.

4.4 Return-Level Estimation Under Parametric GEV Truth

We next examine the finite-sample behavior of GEV return-level estimation under the stationary block-maxima framework. Table 4.2 reports the Monte Carlo mean, bias, and RMSE of $\hat{z}_T$ for $T \in \{10, 50, 100\}$.

The central pattern is again that bias is modest relative to total estimation error. At $n = 95$, the biases for the 10-, 50-, and 100-year return levels are 1,620, 7,338, and 12,977, whereas the corresponding RMSE values are 19,096, 55,870, and 84,048. Thus, as with parameter estimation, finite-sample uncertainty is driven primarily by sampling variability rather than by large persistent bias.

RMSE declines strongly as record length increases, but it also increases systematically with

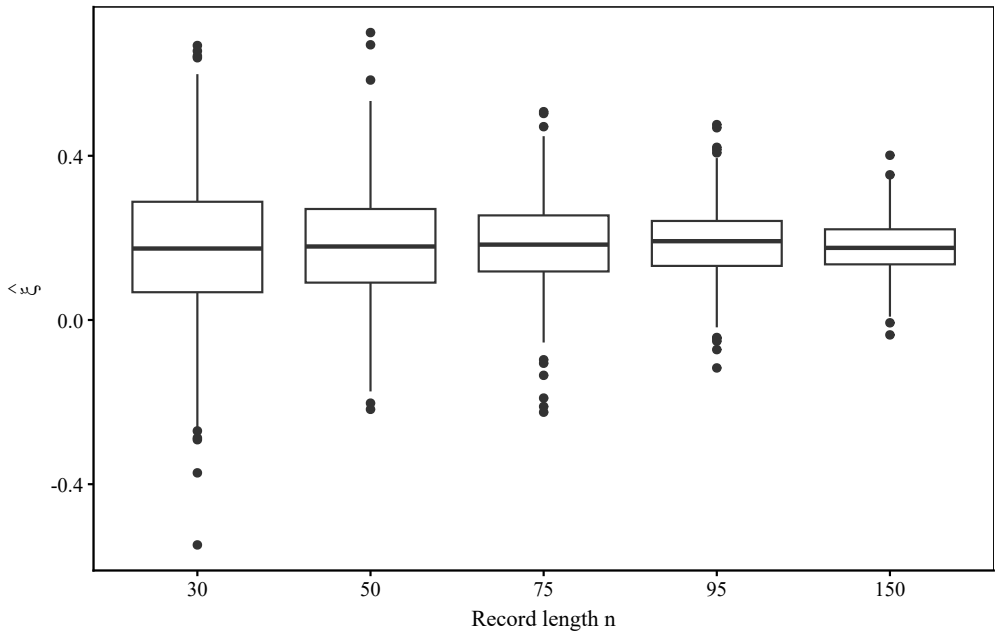

Figure 4.1: Sampling distribution of $\hat{\xi}$ across record lengths under parametric GEV truth.

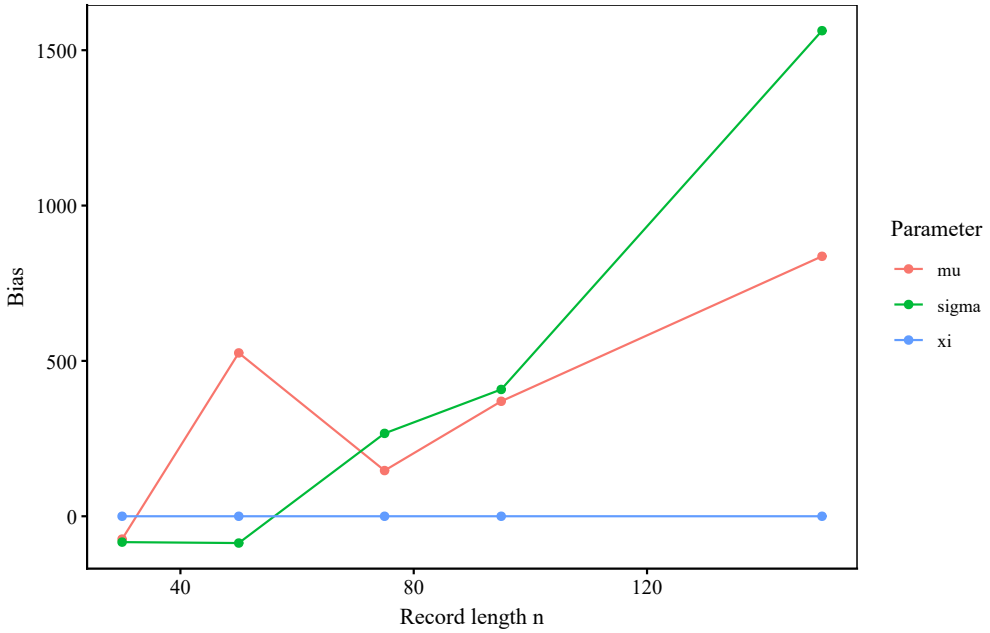

Figure 4.2: Monte Carlo bias of GEV parameter estimators versus record length.

Table 4.2: Monte Carlo mean, bias, and RMSE of return-level estimators under parametric GEV truth.

n	T	Mean($\hat{z}_T$)	Bias	RMSE
30	10	204145.55	2604.67	37919.39
30	50	346994.65	24124.38	176869.96
30	100	435182.90	48805.07	346305.33
50	10	202785.96	1245.08	26000.46
50	50	333531.24	10660.97	88597.35
50	100	407497.19	21119.36	141845.97
75	10	203125.54	1584.66	21103.30
75	50	332092.38	9222.10	65129.06
75	100	403004.35	16626.52	99753.16
95	10	203160.91	1620.02	19095.63
95	50	330208.49	7338.21	55870.30
95	100	399354.50	12976.67	84047.89
150	10	201234.35	-306.53	15068.59
150	50	323723.03	852.75	41195.46
150	100	388609.71	2231.88	60229.81

the return period. The first feature indicates that longer records materially improve the stability of return-level estimation. The second reflects the increasing difficulty of extrapolation farther into the upper tail. Figure 4.4 shows both effects clearly: for each fixed return period, RMSE falls with n, while for each fixed n, RMSE is smallest for $T = 10$ and largest for $T = 100$. Increasing record length improves estimation accuracy, but substantial variability remains at long return periods.

Figure 4.3 shows that return-level bias is largest for longer return periods and smaller record lengths, with the most pronounced bias observed for the 100-year return level at small n. For $T = 50$ and $T = 100$, bias decreases substantially as the record length increases, indicating improved stability of estimation with larger samples. In contrast, the bias for the 10-year return level is relatively small across all n and exhibits only modest variation.

At the Potomac record length ($n = 95$), return-level bias is substantially reduced relative to smaller sample sizes and is small compared with the scale of the estimated return levels, although it remains non-negligible for longer return periods.

The full distribution of the 100-year return-level estimator is shown in Figure 4.5. The distribution narrows appreciably with record length, but at shorter records it is also distinctly right-skewed. That feature is consistent with the positive-tail GEV setting found in the real-data analysis. Because high return levels depend nonlinearly on the shape parameter, fluctuations in $\hat{\xi}$ can induce disproportionately large values of $\hat{z}_{100}$. The consequence is a distribution with occasional

large upper realizations when n is small. As the sampling variability of $\hat{\xi}$ contracts, the distribution of $\hat{z}_{100}$ becomes both narrower and less skewed.

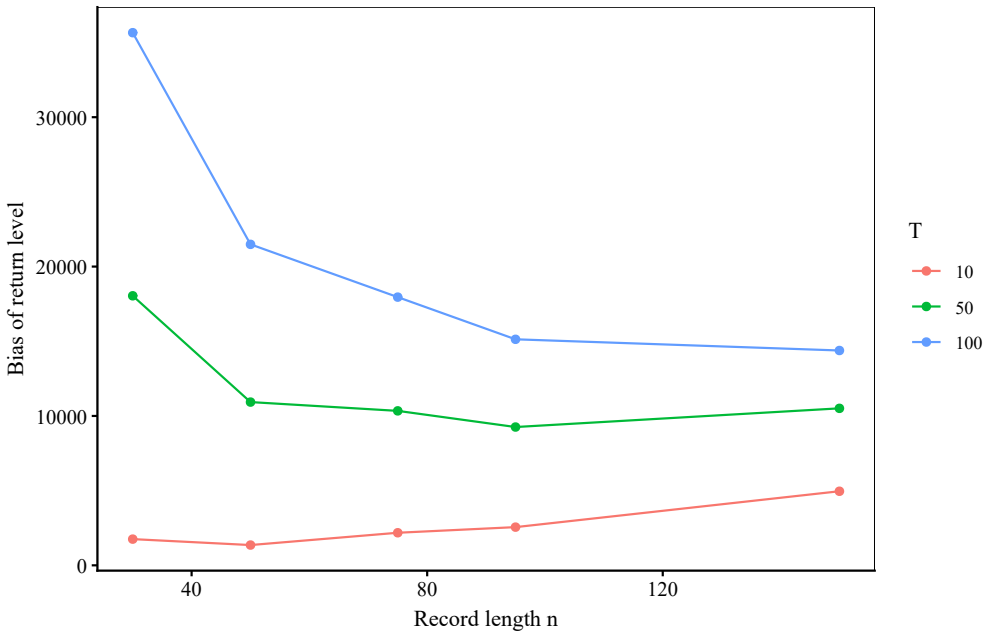

Figure 4.3: Return-level bias versus record length under parametric GEV truth.

4.5 Confidence Interval Performance

We now turn to the performance of the nominal 95% profile-likelihood confidence intervals. Table 4.3 summarizes empirical coverage probability and mean interval width under parametric GEV truth.

Two features stand out. First, empirical coverage is below the nominal 0.95 level for every (n, T) combination shown. Undercoverage is most severe for the 10-year return level, where coverage ranges from 0.730 to 0.818, but it is also present for the 50- and 100-year return levels. Figure 4.6 shows that none of the coverage curves reaches the nominal line over the record-length range studied here. The finite-sample message is therefore necessarily cautious: profile-likelihood intervals provide informative likelihood-based uncertainty summaries, but their nominal 95% level does not guarantee exact 95% coverage at the sample sizes considered here.

Second, interval width behaves in a much more regular manner. Mean width decreases

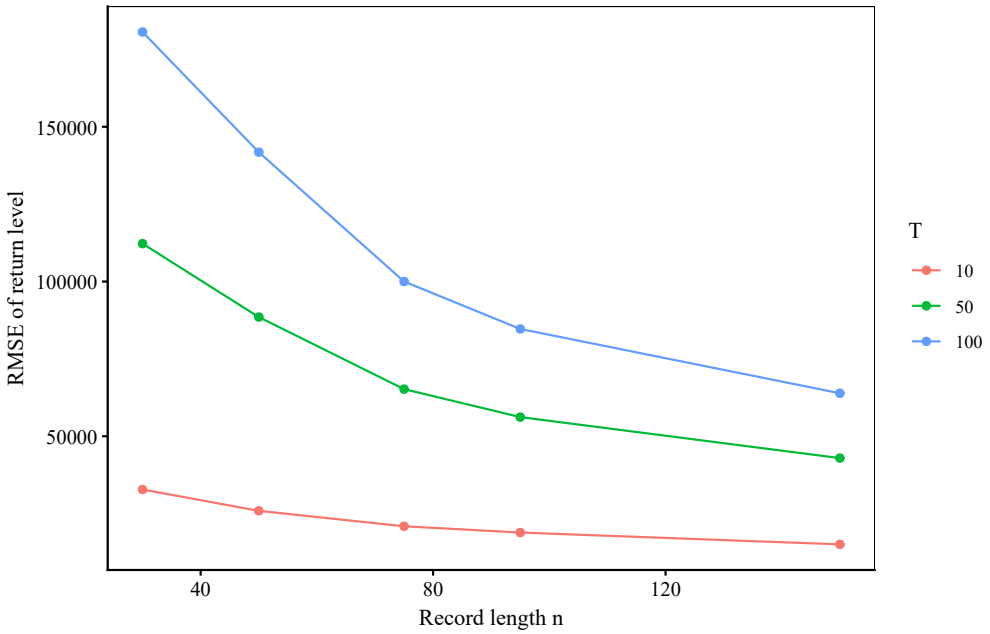

Figure 4.4: Return-level RMSE versus record length under parametric GEV truth.

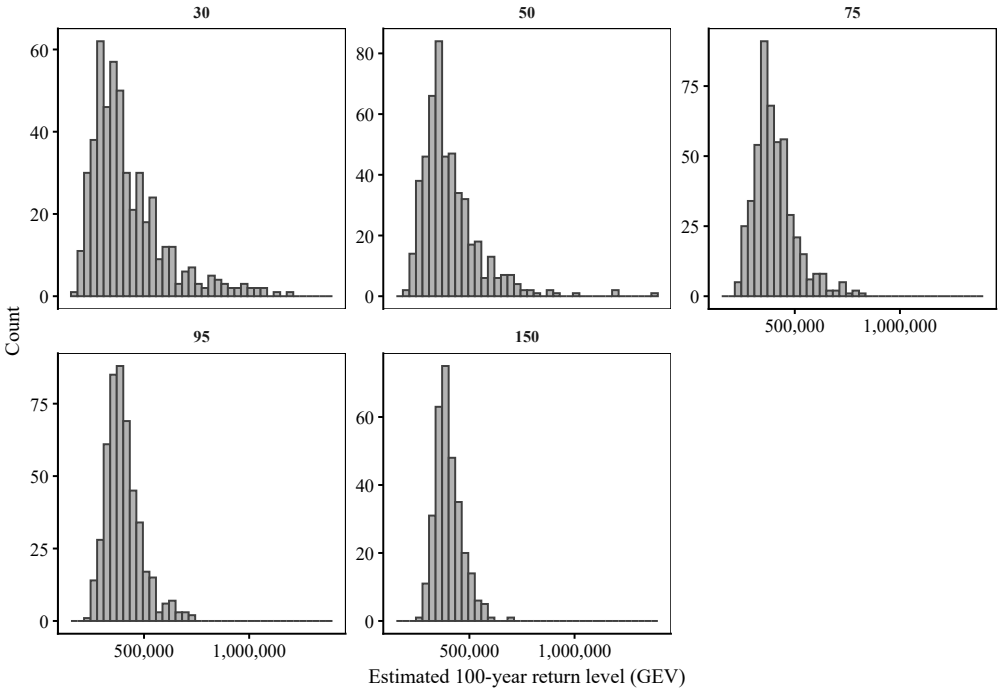

Figure 4.5: Monte Carlo sampling distribution of the 100-year return-level estimator under parametric GEV truth.

Table 4.3: Empirical coverage probability and mean profile-likelihood confidence interval width under parametric GEV truth.

n	T	Coverage	Mean Width
30	10	0.816	116,321
30	50	0.922	390,142
30	100	0.930	419,080
50	10	0.818	74,503
50	50	0.898	310,275
50	100	0.918	402,356
75	10	0.776	56,577
75	50	0.898	240,023
75	100	0.907	354,984
95	10	0.758	47,551
95	50	0.894	197,220
95	100	0.899	304,986
150	10	0.730	35,002
150	50	0.903	140,986
150	100	0.923	217,396

monotonically with record length for all three return periods and increases with the extremity of the return period at every fixed n. Figure 4.7 shows these two patterns clearly. Longer records produce materially more precise intervals, but long-horizon extrapolation remains inherently uncertain. In particular, the 50- and 100-year intervals remain wide even at $n = 95$, which is fully consistent with the empirical Potomac analysis in Chapter 3.

For the Potomac application, the mixed message is important. At $n = 95$, the intervals are much narrower than they would be in shorter records, so the available monitoring history yields a genuine precision advantage. At the same time, the simulation indicates that nominal 95% profile-likelihood intervals can still undercover in moderate samples, even when the annual-maxima model is correctly specified. This is one reason the interpretation of long-horizon return levels should remain appropriately cautious.

4.6 Model Misspecification: GEV Truth versus Gumbel Fit

The real-data analysis in Chapter 3 favored the GEV model under the stationary block-maxima framework over the constrained Gumbel alternative. The parametric simulation allows that preference to be examined more directly by fitting both models to data that are generated under GEV truth.

Table 4.4 shows that the constrained Gumbel fit induces persistent negative bias, especially for high return levels. For the 100-year return level, the Gumbel bias is approximately -1.09×10^5

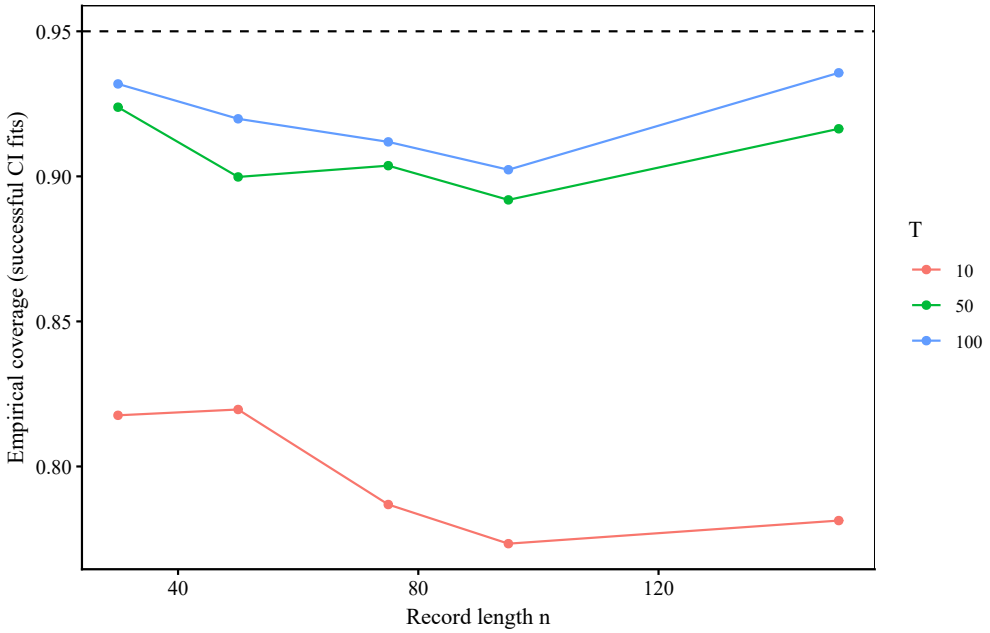

Figure 4.6: Empirical coverage probability versus record length (nominal 95%).

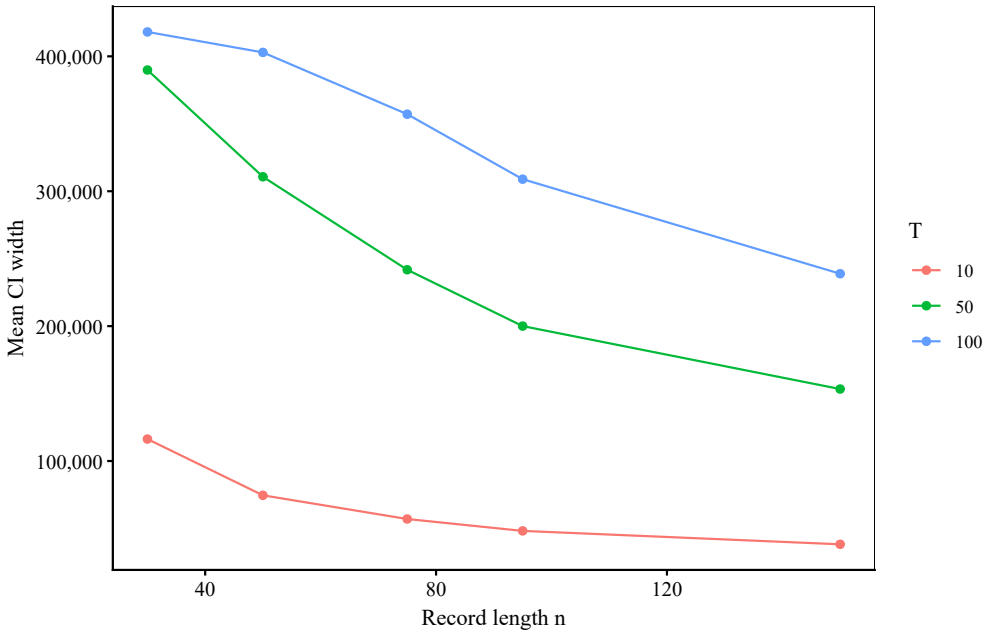

Figure 4.7: Mean profile-likelihood confidence interval width versus record length.

Table 4.4: Return-level bias and RMSE under GEV truth: comparison of GEV and Gumbel fits.

n	T	Bias (GEV)	RMSE (GEV)	Bias (Gumbel)	RMSE (Gumbel)
30	10	2,605	37,919	-22,363	31,337
30	50	24,124	176,870	-75,136	82,362
30	100	48,805	346,305	-109,662	116,318
50	10	1,245	26,000	-22,036	27,957
50	50	10,661	88,597	-75,110	79,549
50	100	21,119	141,846	-109,732	113,771
75	10	1,585	21,103	-21,777	26,083
75	50	9,222	65,129	-74,482	77,663
75	100	16,627	99,753	-109,347	112,303
95	10	1,620	19,096	-21,195	24,715
95	50	7,338	55,870	-73,987	76,607
95	100	12,977	84,048	-108,900	111,316
150	10	-307	15,069	-18,117	20,668
150	50	853	41,195	-75,659	77,459
150	100	2,232	60,230	-109,935	111,544

across the entire record-length grid. The fact that this bias changes very little with n indicates that it is structural rather than finite-sample: when the true shape parameter is positive, fixing $\xi = 0$ systematically suppresses the upper tail and therefore underestimates high quantiles.

The correctly specified GEV fit behaves differently. Its bias at long return periods is positive in short samples but decreases substantially as record length grows. Thus, the GEV bias is primarily a finite-sample phenomenon, while the Gumbel bias is the consequence of forcing the wrong tail form.

The RMSE comparison reveals a bias--variance tradeoff that is especially important in practice. In the smallest samples, the Gumbel fit can appear competitive because it avoids the extra variability associated with estimating ξ. For the 100-year return level at $n = 30$, for example, the GEV fit has much smaller bias than the Gumbel fit but much larger RMSE. As the sample size increases, however, the variance penalty for estimating ξ shrinks, while the Gumbel structural bias remains. By $n = 75$, the GEV fit already has smaller RMSE than the Gumbel fit for the longer return periods, and that advantage is clear at $n = 95$ and $n = 150$.

This comparison provides a direct simulation-based complement to the real-data results of Chapter 3. The positive fitted shape parameter in the Potomac analysis was not merely a formal modeling detail; it materially affects return-level estimation. At a record length comparable to the Potomac series, the GEV model is preferable not only because it reduces structural bias but also because it improves overall RMSE for the return periods that matter most in long-horizon flood assessment.

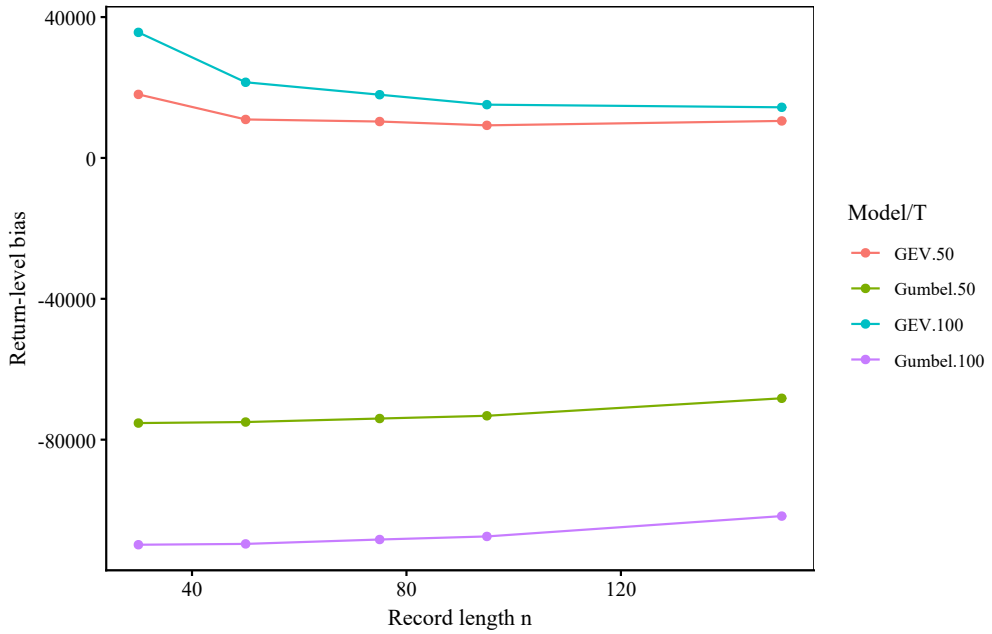

Figure 4.8: Monte Carlo bias of return-level estimators under GEV truth, comparing the correctly specified GEV fit and the constrained Gumbel fit across record lengths.

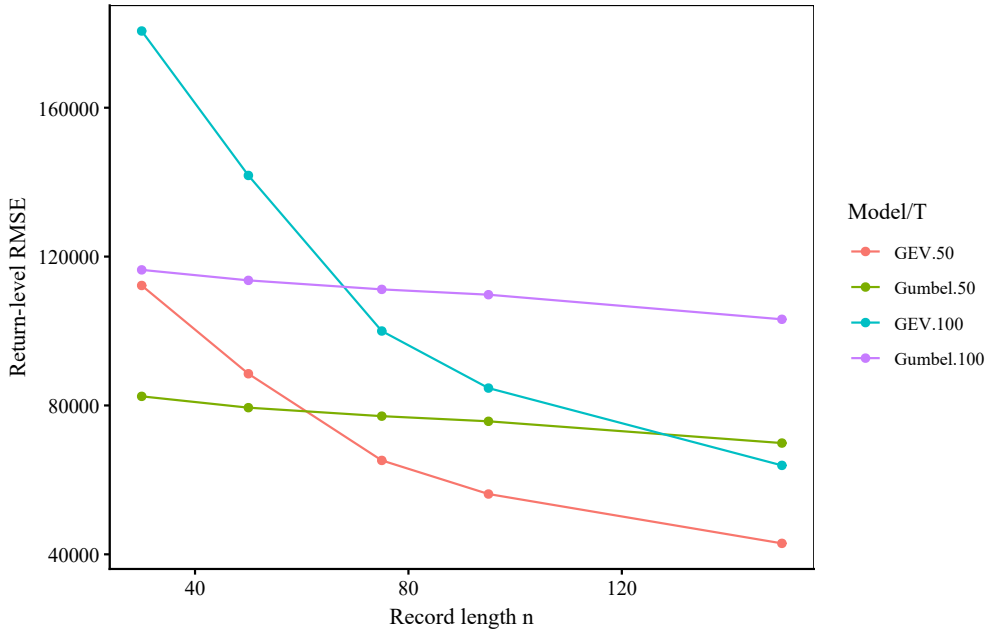

Figure 4.9: Monte Carlo RMSE of return-level estimators under GEV truth, comparing the correctly specified GEV fit and the constrained Gumbel fit across record lengths.

4.7 Robustness via Year-Resampling of Daily Discharge

The final component of the simulation study considers annual maxima generated from a more realistic daily-scale mechanism, focusing on the behavior of return-level estimates and the associated uncertainty measures under this setting. In this experiment, complete calendar years of daily discharge are resampled with replacement, annual maxima are recomputed, and the resulting block maxima are analyzed using the same GEV-based return-level workflow under the stationary block-maxima framework.

The year-resampling design is more realistic than direct GEV simulation because it retains the within-year hydrologic structure present in the observed daily record. For that reason, it is especially informative as a robustness check. At the same time, it does not specify an exact parametric annual-maxima truth, so the reported coverages are best interpreted as reference-based diagnostics relative to the same GEV-calibrated target used in the parametric experiment.

Table 4.5: Year-resampling robustness: empirical coverage and mean 95% CI width.

n	T	Coverage	Mean Width
30	10	0.732	91,271
30	50	0.870	343,790
30	100	0.890	397,920
50	10	0.722	61,250
50	50	0.870	259,342
50	100	0.888	353,517
75	10	0.704	46,808
75	50	0.868	194,981
75	100	0.880	294,481
95	10	0.682	40,845
95	50	0.848	167,324
95	100	0.871	258,674
150	10	0.720	31,838
150	50	0.889	128,046
150	100	0.906	196,773

Table 4.5 shows that the precision pattern from the parametric experiment is preserved. Mean interval width decreases monotonically with record length for all three return periods, and the reduction is substantial. The longer the effective record, the more concentrated the resulting return-level inference becomes.

Coverage, however, remains below the nominal 0.95 level throughout. Indeed, the year-resampling experiment generally produces weaker reference-based coverage than the idealized parametric benchmark. At the Potomac record length $n = 95$, coverage under parametric GEV truth

is $(0.758, 0.894, 0.899)$ for $T = (10, 50, 100)$, whereas the corresponding year-resampling values are $(0.682, 0.848, 0.871)$. Thus, realistic within-year hydrologic structure does not eliminate the undercoverage seen in the parametric experiment; if anything, it accentuates it.

The graphical summaries reinforce this interpretation. Figure 4.10 shows that all coverage curves remain below the nominal line. Figure 4.11 shows the steady decrease in interval width with increasing record length. Figure 4.12 shows that the sampling distribution of the estimated 100-year return level remains right-skewed under year-resampling as well, although dispersion narrows as n increases. Compared with the parametric GEV experiment, the distributions under year-resampling are generally more concentrated, with less pronounced extreme right-tail realizations, while preserving the same qualitative right-skewed shape.

The practical implication is that the stationary block-maxima framework with a GEV model remains useful for point estimation and for relative comparisons of uncertainty, but the calibration of likelihood-based intervals can be sensitive to finite-sample effects and to how the annual maxima are generated. This reinforces that likelihood-based intervals should be interpreted as informative uncertainty summaries rather than exact finite-sample guarantees, particularly at longer return periods.

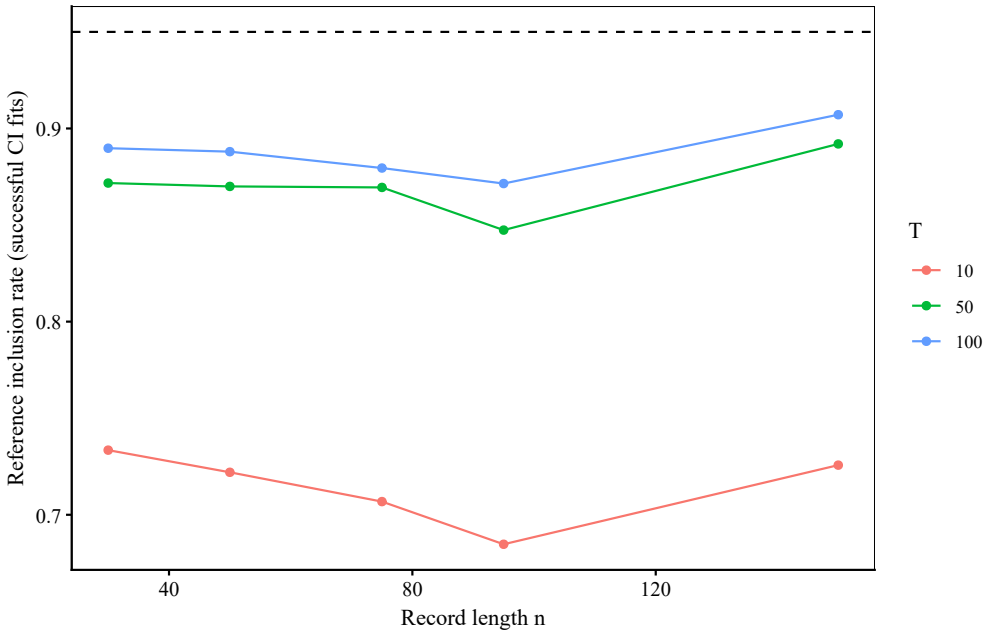

Figure 4.10: Year-resampling: empirical coverage versus record length.

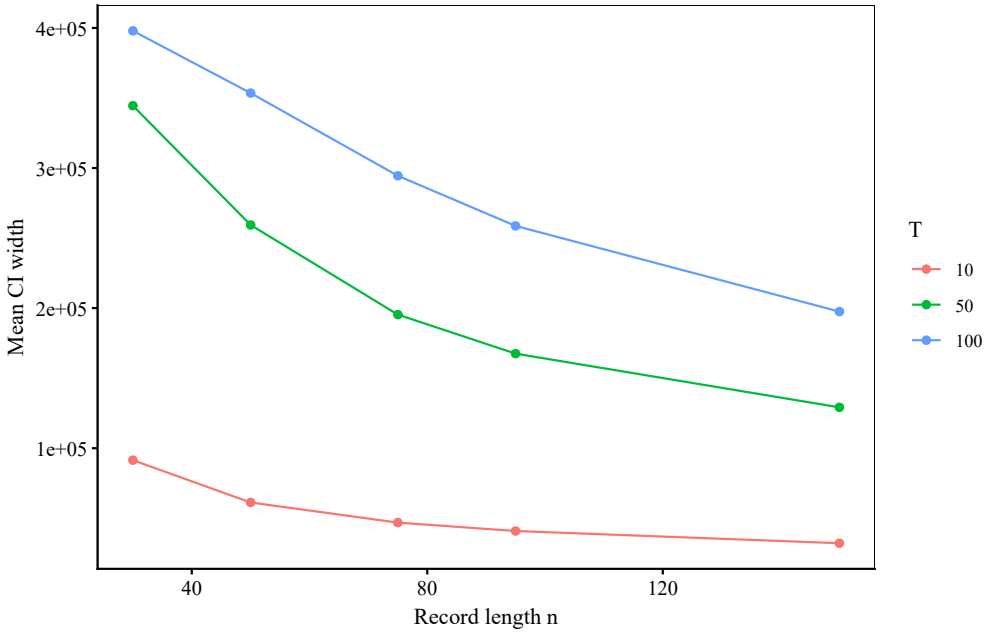

Figure 4.11: Year-resampling: mean 95% confidence interval width versus record length.

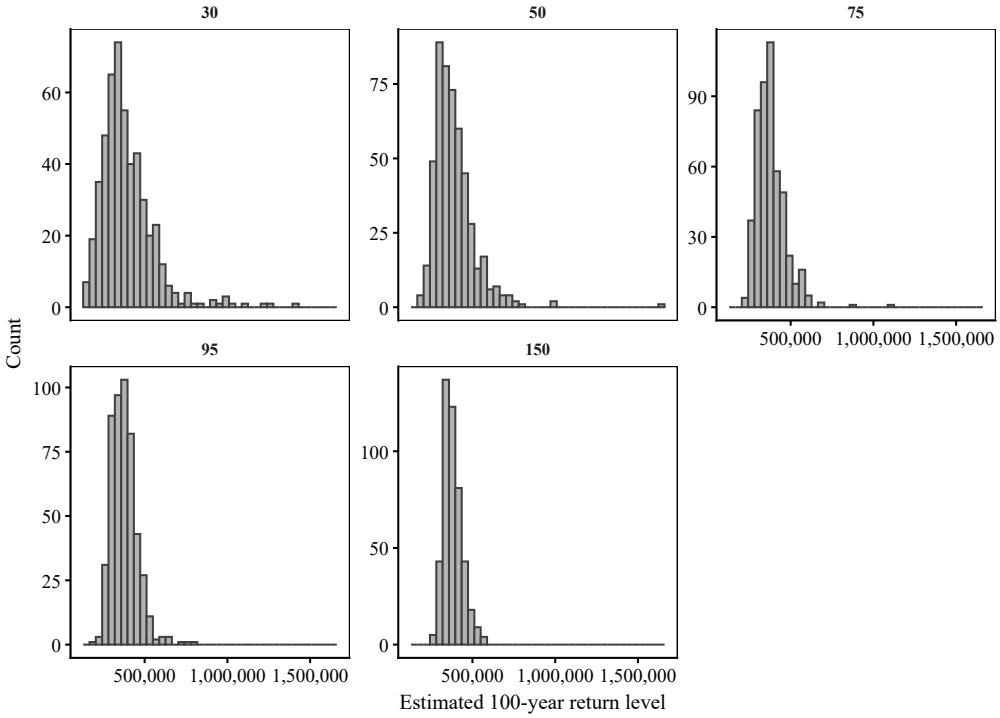

Figure 4.12: Year-resampling: sampling distribution of estimated 100-year return levels.

4.8 Chapter Conclusion

This chapter evaluated the finite-sample reliability of the GEV-based inferential framework under the stationary block-maxima assumption used in Chapter 3. The main conclusion is balanced. At record lengths comparable to the Potomac series, the framework is not so unstable as to make inference unusable. Parameter estimation and return-level estimation are materially more reliable at $n = 95$ than they are in much shorter records, and the gains from additional record length are clear throughout the simulation study.

At the same time, the simulations show that moderate-sample reliability is not the same as negligible uncertainty. Even under correct parametric GEV truth, long-horizon return levels remain variable, the sampling distribution of the 100-year estimator is right-skewed in shorter records, and nominal 95% profile-likelihood intervals tend to undercover. The year-resampling experiment reaches the same broad conclusion while adding further caution: uncertainty quantification is somewhat weaker when annual maxima are generated from a more realistic daily-scale hydrologic structure.

The model-misspecification study also strengthens the substantive conclusions of the empirical chapter. When the true shape parameter is positive, fitting a constrained Gumbel model introduces persistent negative bias in high return-level estimation. Although that misspecified model can appear competitive in very small samples because it reduces variance, its structural bias dominates once the record length reaches the moderate range relevant here. For the Potomac application, this supports the preference for the stationary GEV model over the nested Gumbel alternative.

Overall, the simulation evidence supports the use of a stationary block-maxima framework with a GEV model for the Potomac River annual maxima while also highlighting important limitations, particularly regarding finite-sample uncertainty and the calibration of profile-likelihood intervals. The observed record length is sufficient to yield meaningful and interpretable inference, but statements about very long return periods and nominal profile-likelihood coverage should still be interpreted with appropriate finite-sample caution.

All analyses and figures in Chapters 3 and 4 were produced using R. The code used to generate these results is available from the author upon request.

Chapter 5

Conclusion

This project examined stationary extreme-value inference for annual maximum river discharge through a combined empirical and simulation-based study. The empirical component analyzed annual maximum daily discharge for the Potomac River at Little Falls under stationary generalized extreme value (GEV) and Gumbel models, while the simulation component evaluated how the same likelihood-based inferential framework behaves at finite record lengths. Taken together, these two parts provide a unified assessment of stationarity, model choice, return-level estimation, and uncertainty quantification for hydrologic extremes.

The analysis was motivated by three related questions. First, is a stationary annual-block framework reasonable for the Potomac annual maxima? Second, do the data support a nonzero tail parameter, so that the unrestricted GEV model is preferred to the nested Gumbel specification? Third, at record lengths relevant to hydrologic applications, how reliable are the resulting point estimates and profile-likelihood uncertainty summaries? The preceding chapters addressed these questions empirically for the Potomac record and methodologically through simulation.

5.1 Summary of Main Findings

The results of Chapter 3 support the use of a stationary annual-block framework for the Potomac annual maxima. The annual-maxima series shows substantial interannual variability, but neither the graphical trend assessment nor the Mann--Kendall test provided evidence of a monotone temporal trend. This conclusion was reinforced by the model-based comparison between a stationary

GEV specification and a nonstationary GEV specification with linear trend in location, which likewise provided no evidence that a nonstationary location model improved the fit. Thus, while this does not rule out every possible form of nonstationarity, it does support the stationary annual-block framework as a reasonable working basis for inference in this dataset.

Within that stationary framework, the empirical analysis consistently favored the unrestricted GEV model over the constrained Gumbel alternative. The estimated shape parameter was positive, its likelihood-based interval excluded zero, the Q--Q diagnostics showed better agreement for the GEV fit in the upper tail, and the likelihood ratio test rejected the Gumbel restriction. These findings indicate that the difference between the two models is not a minor technical refinement, but a substantive difference in how the upper tail of annual maximum discharge is represented.

This difference becomes most visible in the return-level analysis. For shorter return periods, the fitted GEV and Gumbel models produce broadly similar point estimates, which is unsurprising given that both models fit the central portion of the annual-maxima distribution reasonably well. As the return period increases, however, the two models diverge substantially. The GEV fit produces larger return levels and wider profile-likelihood intervals, a pattern that is consistent with the positive estimated shape parameter and the heavier upper tail implied by that fit. The profile-likelihood plots make clear that these wider intervals are not an incidental numerical feature, but arise from the flatter curvature of the GEV profile in the return-level parameterization, especially at longer return periods. The practical implication is that conclusions about extreme flood magnitudes depend materially on tail specification, particularly when the return period extends well beyond the length of the observed record.

The simulation study in Chapter 4 clarifies how these empirical findings should be interpreted at finite sample sizes. Under parametric GEV truth, maximum likelihood estimation of the GEV parameters exhibits small Monte Carlo bias relative to root mean squared error (RMSE), so total estimation error is driven primarily by sampling variability rather than by persistent systematic bias. For both the model parameters and the return levels, RMSE decreases as record length increases, which is consistent with the broad qualitative behavior predicted by likelihood asymptotics. At the same time, return-level estimation becomes more difficult as the return period grows, because longer-horizon quantiles depend more strongly on the upper tail of the fitted model and therefore inherit amplified parameter uncertainty.

The interval results are more cautious than the point-estimation results. Although profile-likelihood

intervals become substantially narrower as the number of annual maxima increases, the simulation study shows that nominal 95% profile-likelihood intervals can still undercover in moderate samples. This pattern appears under the idealized parametric GEV design and remains visible under the year-resampling experiment. Accordingly, the confidence intervals reported in the empirical chapter are informative and useful, but they should be interpreted as finite-sample likelihood-based uncertainty summaries rather than as exact calibration guarantees.

The misspecification experiment further strengthens the case for the GEV model. When data are generated from a GEV distribution with nonzero shape parameter, fitting a Gumbel model induces persistent structural underestimation of high return levels. In very short samples, the constrained Gumbel fit can sometimes appear competitive because it avoids the extra variance associated with estimating the shape parameter. That advantage does not persist at moderate record lengths. For record lengths comparable to the Potomac series, the correctly specified GEV model is preferable because it avoids the large structural bias introduced by forcing the tail shape to zero.

Finally, the year-resampling study shows that the main qualitative conclusions are not merely artifacts of an idealized parametric design. When complete calendar years of daily discharge are resampled and annual maxima are recomputed, longer records still produce more stable return-level estimates and narrower intervals. However, reference-based interval calibration is generally weaker than in the parametric experiment. This reinforces the broader conclusion that realistic hydrologic structure matters most for uncertainty quantification rather than for the overall direction of the point-estimation results.

5.2 Interpretation and Practical Implications

Several conclusions emerge from the project as a whole. First, tail-shape specification is practically consequential in flood-frequency analysis. The choice between the GEV model and its Gumbel submodel affects not only the fitted tail behavior in an abstract statistical sense, but also the resulting magnitude and uncertainty of long-horizon return levels. In applications where long-horizon extrapolation is central, tail specification should therefore be viewed as a substantive modeling decision rather than as a secondary modeling detail.

Second, the Potomac record length of 95 annual maxima appears to lie in a moderate-sample regime rather than in an extreme small-sample regime. The simulation study shows clear improvement

in estimation stability when moving from short records to record lengths near 95 years. This is encouraging for the observed-data analysis. At the same time, moderate sample size does not imply negligible uncertainty. For long return periods such as 100 years, the combination of finite sample size and upper-tail extrapolation still produces substantial variability and wide confidence intervals, so return-level statements must be interpreted with appropriate care.

Third, the project illustrates the value of pairing empirical modeling with a simulation study tailored to the fitted application. The empirical analysis identifies a preferred model and produces site-specific return-level estimates, while the simulation study clarifies how much confidence can reasonably be placed in those estimates at the available record length. This combined approach yields a more transparent interpretation than either component could provide on its own. In particular, it allows the Potomac results to be read not only as fitted outputs from a single dataset, but as part of a broader finite-sample picture that explains how those outputs are likely to behave under repeated sampling.

5.3 Limitations and Directions for Future Work

The conclusions of this project should be read together with several important limitations. The analysis was conducted under a stationary block-maxima framework. Although the observed annual maxima show no evidence of a monotone trend and the model-based comparison did not support a linear trend in location, the possibility of more subtle nonstationarity, regime change, or covariate effects cannot be ruled out. Future work could extend the analysis to nonstationary GEV models with time-varying or covariate-dependent parameters, allowing direct assessment of how climate variability, watershed conditions, or other external drivers may influence extreme discharge behavior.

A second limitation is inherent to the block-maxima approach itself. Reducing each year to a single annual maximum provides a clean and interpretable framework for return-level analysis, but it also discards potentially relevant information from other large events within the same year. Peaks-over-threshold methods offer a natural complementary direction because they can use more of the extreme-data information and may improve statistical efficiency when a suitable threshold model is available. A direct comparison between block-maxima and threshold-based approaches would therefore be a useful extension of the present work.

A third limitation concerns the robustness experiment. Resampling complete calendar years preserves realistic within-year dependence and seasonality, but it treats years as independent units. Any long-range interannual dependence or low-frequency hydroclimatic structure is therefore not explicitly modeled. Additional work on resampling schemes or hierarchical models that better reflect cross-year dependence could help refine uncertainty assessment for environmental extremes.

Methodologically, several extensions are also worth pursuing. One is a more detailed study of finite-sample coverage behavior for profile-likelihood intervals across a broader range of tail parameters and sample sizes. Another is the use of Bayesian extreme-value models, which could incorporate parameter uncertainty and model uncertainty within a single probabilistic framework. Finally, applying the same empirical-and-simulation strategy to other river systems would help determine how general the present findings are across contrasting hydrologic regimes.

5.4 Final Remarks

This project has shown that modeling annual maxima with a GEV distribution under a stationary block-maxima framework provides a statistically coherent and practically informative approach for analyzing extreme discharge at the Potomac River at Little Falls. The empirical analysis supports a heavier-tailed GEV specification over the nested Gumbel alternative, and the simulation study shows that inference at the observed record length is substantially more stable than it would be for much shorter hydrologic records. At the same time, the project also makes clear that finite-sample variability remains an essential part of extreme-value analysis, especially for long return periods and for the interpretation of nominal confidence intervals.

More broadly, the study highlights a central principle of hydrologic extreme-value modeling: inference for high quantiles is shaped jointly by the tail model, the available record length, and the way uncertainty is quantified. Careful likelihood-based analysis, combined with problem-specific simulation, provides a defensible way to make those features explicit. In that sense, the main contribution of this project is not only the fitted Potomac results themselves, but also the clearer understanding of how those results should be read, qualified, and applied in practice.

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